Phase-Modulated Compound Sparsity Severity in 7B-Scale Mathematical Reasoning Models

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Abstract

Compound sparsity—simultaneously pruning attention heads and evicting KV cache entries—is essential for efficient LLM inference but produces super-additive quality degradation whose structure is poorly understood. We find that compound sparsity severity is not a fixed constant but an emergent property of the interaction among eviction strategy, reasoning phase, and sparsity level. On DeepSeek-R1-Distill-Qwen-7B (MATH-500), importance-based eviction exhibits a monotonically increasing phase–severity gradient: the paired median difference Δ rises from 0.39 at sparsity s=0.15 to 8.48 at s=0.30 (Spearman ρ=1.0; all Bonferroni-corrected $p < 0.005$). Random eviction develops a detectable gradient only above $s \approx 0.20$, at 2.7–17$\times$ smaller magnitude. A scoring-independent control (recency eviction) produces no positive phase gradient: Δ vanishes at $s \leq 0.25$ and remains non-significant at s=0.30, confirming that the interaction is attention-score-specific rather than an inherent property of token removal. The pattern replicates on Qwen2.5-7B-Instruct (Δ: 0.07 $\rightarrow$ 3.16; ρ=1.0), shows consistent phase gradient direction on GSM8K, and generalizes to LlamaForCausalLM (n=250; d_z=0.81–1.15, all $p < 10^{-29}$). Per-phase Gini analysis identifies *importance discrimination collapse*—the degeneration of head importance scores toward uniformity under compound stress—as a correlational diagnostic of this severity divergence. Prior work characterizes compound degradation as phase-agnostic; we show that phase modulates severity by 15–17$\times$ on DeepSeek and 2.7–3.5$\times$ on Qwen, specifically driven by attention-score-based selection. These findings establish phase-gated analysis as a principled diagnostic framework with direct implications for phase-aware budget allocation.

1 Introduction

Long-context language models place acute pressure on the key–value (KV) cache, which grows linearly with sequence length (Kwon et al., 2023; Pope et al., 2023). Two complementary responses—*attention head pruning* (Voita et al., 2019; Michel et al., 2019) and *KV cache eviction* (Li et al., 2024; Feng et al., 2025)—are increasingly deployed simultaneously as *compound sparsity*, but the resulting quality degradation is often super-additive (Liao et al., 2026; Mozaffari et al., 2025). Recent evaluations show that reasoning capabilities degrade disproportionately under compound compression (von Rad et al., 2026), and that per-dimension safe zones intersect to form a smaller joint safe region (Ma et al., 2026).

Prior work treats compound interaction magnitude as roughly constant across generation (Liao et al., 2026; von Rad et al., 2026), despite evidence that pruning sensitivity depends on representational phase (Shi et al., 2026) and that no single sparse attention strategy dominates across generation stages (Nawrot et al., 2025). This assumption is problematic for reasoning models, whose chain-of-thought generations (Wei et al., 2022) alternate between broadly distributed attention (exploratory reasoning) and sharply concentrated attention (structured computation) (Ramachandran et al., 2026). Phase-aware methods like ThinKV (Ramachandran et al., 2026) apply different KV strategies per phase but without considering compound interaction, while importance-based eviction itself faces challenges from score collapse (Choi et al., 2025), greedy bias (Sood et al., 2026), and temporal dynamics (Zeng et al., 2026). These methods each optimize a single pruning axis; we characterize the interaction when multiple axes are composed. If compound severity varies across these phases, a uniform sparsity budget may be safe in one phase yet destructive in another.

We find that compound sparsity severity is an emergent property of the three-way interaction among eviction strategy, reasoning phase, and sparsity level. Under importance-based KV cache evic-

tion, the Interaction Index—a measure of super-additive degradation—concentrates in low-sparsity reasoning phases and is largely absent in high-sparsity phases. On DeepSeek-R1-Distill-Qwen-7B (MATH-500), the paired median difference rises from 0.39 at s=0.15 to 8.48 at s=0.30 (Spearman ρ=1.0; all Bonferroni-corrected $p < 0.005$). The pattern replicates on Qwen2.5-7B-Instruct (Δ: 0.07 $\rightarrow$ 3.16; all $p < 0.005$), with cross-architecture support from DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, n=250), where the importance phase gradient replicates with large effect sizes (d_z=0.81–1.15; all $p < 10^{-29}$) despite reduced strategy specificity (2.0–2.6$\times$ vs. 15–17$\times$ on Qwen2-family models). Random eviction develops a detectable phase gradient only above $s \approx 0.20$, at 2.7–17$\times$ smaller magnitude than importance-based (where both strategies reach significance at $s \geq 0.25$), supporting that the dominant gradient is strategy-specific. A scoring-independent control (recency eviction, which removes the oldest positions without attention scores) produces no phase gradient ($\Delta = -0.003$, CI spans zero), ruling out segmentation artifact and confirming that the interaction is specifically driven by attention-score-based selection (§5.6). Cross-benchmark validation on GSM8K reveals consistent phase gradient direction: DeepSeek replicates the monotonic dose-response (Δ: 0.24 $\rightarrow$ 4.68; all $p \leq 0.0003$), while Qwen shows directional consistency at $s \geq 0.25$ but is inconclusive due to generation truncation (§5.8). Strategy specificity attenuates on GSM8K relative to MATH-500, indicating that phase modulation generalizes but its magnitude is task-dependent.

Per-phase Gini analysis identifies *importance discrimination collapse*—the degeneration of head importance scores toward uniformity under compound stress—as a correlational diagnostic of this phase–strategy divergence. Both strategies exhibit consistent Gini drops across all tested conditions, yet severity diverges by 2.7–17$\times$, implicating the structural specificity of importance-based pruning as the key differentiator.

Beyond individual findings, we contribute a *phase-gated analysis framework* that decomposes compound severity by reasoning phase and eviction strategy, transforming a scalar interaction metric into a structured diagnostic. This framework—combining the Interaction Index with attention-sparsity-based phase segmentation—reveals structure in compound degradation that aggregate metrics miss, and provides practitioners with a principled basis for phase-aware sparsity allocation.

Our contributions:

1. We characterize a **phase-modulated compound severity gradient**: importance-based eviction exhibits a monotonic dose-response (ρ=1.0; Δ: 0.39 $\rightarrow$ 8.48) with a phase gradient 2.7–17$\times$ larger than random eviction where both strategies are significant ($s \geq 0.25$) on MATH-500; the gradient direction is consistent on GSM8K, while strategy specificity attenuates (§5.8).
2. We replicate the pattern **across training paradigms and architectures**: within-architecture replication on Qwen2.5-7B-Instruct (same Qwen2 backbone, instruction-tuned; Δ: 0.07 $\rightarrow$ 3.16; ρ=1.0) and cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM; n=250; importance phase gradient d_z=0.81–1.15, all $p < 10^{-29}$; strategy specificity reduced to 2.0–2.6$\times$, architecture-dependent; §5.5).
3. We provide a **scoring-independent control**: recency eviction (position-only, no attention scores) under identical compound sparsity produces no phase gradient ($\Delta \approx 0$, CI spans zero), ruling out segmentation artifact and establishing that phase-dependent severity is specifically driven by attention-score-based selection (§5.6).
4. We identify **importance discrimination collapse** via per-phase Gini analysis as a correlational diagnostic: consistent collapse across models and strategies, with the structural specificity of importance-based pruning corresponding to the severity divergence (§6).
5. We validate the phase gradient on **GSM8K** for both models, finding that phase modulation is benchmark-robust while strategy specificity is task-dependent (§5.8).

2 Related Work

Compound compression. Combining multiple compression axes produces non-additive errors. (Liao et al., 2026) report super-additive degradation when combining channel pruning with token eviction; (Mozaffari et al., 2025) formalize compound error for quantization and sparsity. (von Rad et al., 2026) evaluate compound compression across 40+ benchmarks, revealing that reasoning capabilities degrade disproportionately. (Ma et al., 2026) characterize independent phase transition points per compression dimension, implying the joint safe

region is the intersection of per-dimension safe zones. GRASPrune (Wang et al., 2026) learns global budget-constrained gates for joint pruning but treats decisions as static across generation. PM-KVQ (Liu et al., 2026b) shows that progressive quantization—gradually reducing bit-width over the generation process—mitigates cumulative error in long-CoT reasoning, implicitly introducing a temporal dimension to compression severity that resonates with our finding that compound interaction is phase-dependent.

(Kim et al., 2026) show that compression order (prune-then-quantize vs. quantize-then-prune) substantially affects joint quality, and their Progressive Intensity Hypothesis—that weaker perturbations should precede stronger ones—echoes our dose-response finding that interaction severity increases monotonically with sparsity.

These works establish compound interaction but share an implicit assumption: interaction magnitude is constant across the generation process.

Structured pruning. Magnitude pruning (Han et al., 2015) and the lottery ticket hypothesis (Frankle and Carlin, 2019) establish that neural networks contain sparse subnetworks preserving full-model performance. Movement pruning (Sanh et al., 2020) extends this to fine-tuning, learning which weights to zero during task adaptation. For attention heads specifically, (Voita et al., 2019) show most heads are prunable without degradation, and (Michel et al., 2019) demonstrate that head importance varies substantially across layers—findings our head pruning axis builds upon. These methods optimize a single pruning axis in isolation; we characterize the interaction when head pruning operates simultaneously with KV cache eviction.

KV cache eviction. StreamingLLM (Xiao et al., 2024b) establishes attention sinks as a foundational mechanism; DuoAttention (Xiao et al., 2024a) classifies heads into retrieval and streaming types, complementing our analysis of importance discrimination collapse (§6.1). Subsequent work improves eviction through per-head adaptive budgets (Feng et al., 2025), observation-window compression (Li et al., 2024), proxy-token strategies (Chen et al., 2024), deferred importance (Liu et al., 2026a), and worst-case analysis (Feng et al., 2026); (Sood et al., 2026) show that single importance metrics exhibit systematic greedy bias. (Choi et al., 2025) independently show that importance discrimination collapses under heavy single-axis pruning; our results extend this to the compound setting, revealing phase-modulated amplification and a strategy crossover (§5.3) absent in single-axis analysis. Recent approaches target reasoning-aware eviction (Du et al., 2025; Su et al., 2026; Dong et al., 2026), temporal adaptivity (Zeng et al., 2026), channel-level pruning (Xu et al., 2025; Liao et al., 2026), and learned budgets (Zhou et al., 2026). Concurrent work (Liu et al., 2025) evaluates single-axis compression on mathematical reasoning; we extend to the compound setting. These methods each optimize a single pruning axis without characterizing compound interactions.

Phase-aware inference and reasoning. ThinKV (Ramachandran et al., 2026) applies different KV strategies per phase but without compound interaction. (Shi et al., 2026) find pruning sensitivity depends on "silent" vs. "decisive" representation phases. (Monjur et al., 2026) show pruning combined with test-time scaling can exceed unpruned baselines; (Nawrot et al., 2025) find no universally optimal sparse attention strategy across generation stages. (Ananthanarayanan et al., 2026) identify a sharp phase transition in single-axis KV cache compression at high sparsity, characterizing the single-dimension failure mode that our compound analysis extends across multiple axes. R-KV (Cai et al., 2025a) exploits reasoning-specific redundancy for aggressive cache compression, targeting per-axis optimization rather than characterizing compound degradation. Test-time compute scaling (Snell et al., 2024) shows that allocating more inference-time computation can substitute for model scale; this emphasizes the importance of preserving reasoning-phase fidelity, where our work shows compound sparsity is most destructive. ScissorHands (Liu et al., 2023) leverages the persistence of importance hypothesis—that important tokens remain important across generation steps—a property our Gini analysis shows breaks down under compound stress. These works establish phase-dependent compression tolerance along single axes.

Conditional computation. Mixture-of-Experts architectures (Shazeer et al., 2017; Fedus et al., 2022) achieve dynamic sparsity by routing tokens to a subset of parameters, providing a complementary paradigm to post-hoc pruning. MoE sparsity is *designed* (experts are trained to specialize), whereas compound sparsity is *imposed* (pruning removes capacity without retraining). Our phase-

strategy interaction findings suggest that designed sparsity and imposed sparsity may interact differently across reasoning phases—a connection we leave to future work.

Positioning. Our work bridges the compound compression and phase-aware literatures by quantifying how compound severity varies across phases and eviction strategies. Key differentiators: SparK (Liao et al., 2026) characterizes compound interaction magnitude but treats it as phase-invariant; ThinKV (Ramachandran et al., 2026) adapts per phase but considers only single-axis KV compression without compound interaction; HIES (Choi et al., 2025) identifies importance score collapse under single-axis pruning but does not quantify its role in compound settings or its phase dependence. Our strategy specificity finding—that random eviction produces near-zero phase gradients while importance-based eviction exhibits 15–17× amplification—is non-obvious: the *a priori* expectation is that any eviction strategy operating on phase-structured attention should produce some phase gradient, yet random eviction yields effectively none at $s \leq 0.20$. The quantitative dose-response characterization provides target operating points for phase-aware compression controllers.

3 Background and Problem Setup

We formalize compound sparsity, define the Interaction Index, and introduce phase segmentation. Full mathematical details appear in Appendix A.

Compound sparsity. Let $s \in (0, 1)$ denote a unified sparsity budget. *Attention head pruning* removes the bottom-s fraction of heads per layer by Frobenius norm importance (Voita et al., 2019; Michel et al., 2019). *KV cache eviction* removes the bottom-s fraction of token positions from the cache. We compare two strategies: *importance-based*, which scores positions by position-normalized cumulative attention mass (Eq. 5 in Appendix A), and *random*, which evicts uniformly at random (Chen et al., 2024). Applying both simultaneously yields *compound sparsity*; the two axes interact because head pruning alters the importance ranking used for eviction (Liao et al., 2026). Existing joint-pruning approaches such as GRASPrune (Wang et al., 2026) learn global budget-constrained gates but treat decisions as static across generation.

Interaction Index. For a text segment, let PPL_c denote perplexity under condition $c \in \{d, a, k, b\}$ (dense, attention-only, KV-only, compound). Normalizing by the dense baseline yields $r_c = \text{PPL}_c/\text{PPL}_d$, and:

$$\text{II} = r_b - r_a - r_k + 1. \tag{1}$$

$\text{II} > 0$ indicates super-additive degradation; $\text{II} = 0$ indicates additive independence. We adopt the additive null as primary: it assumes $r_b = r_a + r_k - 1$ (compound loss equals sum of individual losses), the most conservative independence assumption. Although the two axes are mechanistically linked (§6), rejection ($\text{II} > 0$) establishes super-additivity regardless of null specification. Appendix B reports parallel results under the multiplicative null ($r_b = r_a \cdot r_k$); the phase-severity gradient is preserved ($G_\times$=21–64×), supporting null-choice invariance. The II distribution is heavily right-skewed; we report medians throughout.

Severity gradient. To capture phase variation, the per-problem *amplification ratio* is:

$$G^{(i)} = \frac{\text{II}^{(i)}_{low}}{\text{II}^{(i)}_{high}}, \tag{2}$$

where $\text{II}^{(i)}_{low/high}$ is the interaction index in the low/high-sparsity phase for problem i. We report $G = \widetilde{G^{(i)}}$, the median across problems, with bootstrap CIs. We use the median of per-problem ratios (not the ratio of medians) to preserve per-problem pairing; see Appendix D for comparison. G serves as an auxiliary descriptive ratio that illustrates the degree of phase concentration; the primary statistical evidence for the severity gradient rests on Δ and paired Wilcoxon tests (§5.2).

Phase segmentation. Inspired by ThinKV's phase-aware perspective on reasoning-time compression (Ramachandran et al., 2026), each generated sequence is partitioned into three phases (low-, mid-, high-sparsity) by tercile boundaries of per-position attention sparsity, using a quantile-based segmentation rather than explicit reasoning delimiters (e.g., `<think>` tokens). Low-sparsity phases correspond to broadly distributed attention (exploratory reasoning); high-sparsity phases to concentrated attention (structured computation). We note that this segmentation is an *operationalization choice*: attention sparsity serves as a continuous proxy for reasoning phase rather than a theoretically grounded boundary. We validate robustness to segmentation granularity (tercile/quartile/quintile; Appendix R) and minimum

phase length (Appendix Q). Validation against DeepSeek's explicit <think> markers using adaptive per-sequence quantile segmentation yields substantial agreement (κ=0.88, AUC-ROC=0.98; n=48; Appendix U). Importantly, phases reflect genuine computational mode differences even under the dense (uncompressed) baseline: per-phase dense perplexity increases monotonically from high (1.10) through mid (1.17) to low (1.18), confirming that low-sparsity phases correspond to inherently harder computation; the phase gradient in compound severity (§5) thus amplifies a pre-existing computational asymmetry rather than creating one. Details in Appendix A.

4 Experimental Setup

Models. We evaluate on two 7B-parameter models sharing the Qwen2 backbone (28 layers, 28 heads, 4-group GQA): **DeepSeek-R1-Distill-Qwen-7B** (DeepSeek-AI, 2025), distilled from DeepSeek-R1 with explicit chain-of-thought traces yielding pronounced phase structure; and **Qwen2.5-7B-Instruct** (Yang et al., 2024), with standard instruction tuning, providing a cross-training-paradigm replication target. Both are loaded in `bfloat16` with eager attention.[1]

Dataset. We evaluate all 500 problems from MATH-500 (Hendrycks et al., 2021). Each problem is decoded greedily (`max_new_tokens`=2048). Problems with degenerate phase assignments (null low or high phase) are excluded, yielding effective sample sizes of n=496 for DeepSeek (4 excluded) and n=475 for Qwen (25 excluded). Qwen exhibits a higher mid-phase null rate (45.2% vs. 10.2% for DeepSeek) due to lower temporal autocorrelation in its attention sparsity patterns (lag-1 r=0.718 vs. 0.894, d= − 1.68); this does not affect the primary gradient metric $\Delta = \text{II}_{low} - \text{II}_{high}$, which requires only low and high phases. Sensitivity analysis across `min_phase_length` $\in \{5, 10, 20\}$ supports robustness (Appendix Q).

Sparsity configuration. We apply symmetric compound sparsity at $s \in \{0.15, 0.20, 0.25, 0.30\}$: at each level, the head pruning rate p=s (fraction of heads removed per layer) equals the KV cache eviction rate e=s (fraction of positions evicted). We compute perplexity under four conditions: dense, attention-only (p=s, e=0), KV-only (p=0, e=s), and compound (p=s, e=s). We compare *importance-based* eviction (position-normalized cumulative attention score, Eq. 5; BOS always retained) and *random* eviction (uniform). Heads are pruned per layer by Frobenius norm; under random eviction, both heads and positions are selected uniformly.

Phase segmentation and metrics. Each sequence is segmented into three phases using the quantile-based method (§3), with `min_phase_length`=10. We compute segment-level perplexity and II (Eq. 1), reporting medians due to heavy right-skew (mean/median ratios up to 43:1). The amplification ratio G (Eq. 2) is reported in Appendix V. Robustness to segmentation granularity (quartile, quintile; Appendix R) and minimum phase length (5, 10, 20 tokens; Appendix Q) is verified.

Segmentation validation. Quantile-based segmentation shows strong agreement with model-intrinsic <think> boundaries on DeepSeek (Cohen's κ=0.88, token-level accuracy 94.2%, AUC-ROC=0.98; n=48, GSM8K): 57.8% of <think> tokens fall in the low-sparsity phase and 62.7% of answer tokens fall in the high-sparsity phase, confirming that attention-sparsity phases recover genuine reasoning structure without requiring explicit delimiters. Boundary proximity is tight (median 1 token; 81.2% within 10 tokens). Appendix U provides full details.

Statistical analysis. All reported confidence intervals use nonparametric bootstrap (B=10,000, seed=42) (Efron and Tibshirani, 1993). Significance tests use Wilcoxon signed-rank (paired within-model) or Mann-Whitney U (cross-training-paradigm), with Bonferroni correction at family-wise k=4 (per model×strategy combination, 4 sparsity levels testing one dose-response hypothesis): each family tests a single directional question ("does Δ increase with s?") under fixed model and strategy, making k=4 the natural grouping. Benjamini-Hochberg FDR correction across all 44 tests is reported as a robustness check in Appendix P; all Bonferroni-significant results remain significant under BH-FDR. Dose-response monotonicity is supported by the Jonckheere-Terpstra test (z=6.2–33.7, all $p < 10^{-9}$), which uses per-problem values (∼496 per group) rather than aggre-

[1] An earlier `float16` configuration produced sporadic NaN values; `bfloat16` eliminated all occurrences.

Table 1: Median of per-problem paired severity differences (Δ) by eviction strategy and sparsity level on DeepSeek-R1-Distill-Qwen-7B (n=496, MATH-500).

Strategy	s=0.15	s=0.20	s=0.25	s=0.30
Importance	0.39	1.21	3.99	8.48
Random	0.01	0.03	0.26	0.51

gated medians.[2] Scaling from n=50 (profiling) to n=500 narrowed bootstrap CI widths by 2.5–5.9× across all 16 conditions, showing precision gains. The paired median difference Δ is the primary metric; G (amplification ratio, Eq. 2) is reported in Appendix V as an auxiliary illustrative ratio.[3]

5 Results

5.1 Strategy-Dependent Severity

Table 1 reports the median compound severity difference across strategies and sparsity levels. Under importance-based eviction, Δ increases from 0.39 at s=0.15 to 8.48 at s=0.30—a 22× amplification (all Bonferroni-corrected $p \ll 10^{-10}$). Random eviction develops a detectable but weak phase gradient above $s \approx 0.20$: at $s \leq 0.20$, Δ is non-significant (0.01–0.03; CIs span zero); at $s \geq 0.25$, Δ becomes significant (0.26–0.51; $p_{\text{Bonf}} < 0.006$) but 15–17× smaller than under importance-based eviction. Both strategies exhibit monotonic dose-response (Jonckheere-Terpstra z=6.2–33.7, all $p < 10^{-9}$), but importance masking effect sizes are an order of magnitude larger.

5.2 Phase-Modulated Amplification

Decomposing II by reasoning phase reveals that strategy-dependent severity is driven by amplification in the low-sparsity phase. Table 2 reports the per-phase breakdown. Under importance-based eviction, the median II in the low-sparsity phase rises from 0.357 (s=0.15) to 8.456 (s=0.30), while the high-sparsity phase remains near zero (0.003–0.038), yielding paired median differences Δ from 0.39 to 8.48 (all $p \ll 10^{-10}$). Random eviction shows a detectable but much weaker phase gradient at $s \geq 0.25$ (Δ=0.26–0.51; $p_{\text{Bonf}} < 0.006$), 15–17× smaller than importance-based at matched sparsity. Figure 1 visualizes the dose-response relationship for both models. The per-problem amplification ratio G (Eq. 2) provides an auxiliary illustrative perspective and is reported in Appendix V. Under the multiplicative null ($r_b = r_a \cdot r_k$; Appendix B), the phase-severity gradient is preserved ($G_\times$=21–64×); we report additive-null values throughout as the more conservative estimates.

5.3 Phase-Strategy Crossover

The phase-strategy crossover is model-dependent. For DeepSeek (n=496), a strategy ranking reversal emerges at all four sparsity levels: at s=0.25, importance-based eviction yields *lower* II than random in the high-sparsity phase (0.021 vs. 1.113) but *higher* II in the low-sparsity phase (3.971 vs. 1.514—a 2.6× reversal; both $p < 10^{-4}$), with reversal magnitude ranging from 2.2× to 4.0× across $s \in \{0.15, 0.20, 0.25, 0.30\}$. Qwen does *not* exhibit this crossover: importance-based eviction produces lower II than random in both phases (e.g., at s=0.25: high 0.004 vs. 1.108; low 1.053 vs. 1.769). This divergence suggests the crossover arises from model-specific attention head specialization patterns rather than a universal compound-sparsity property. Full crossover analysis appears in Appendix C.

5.4 Replication Across Training Paradigms

DeepSeek-R1-Distill-Qwen-7B (distillation-trained) and Qwen2.5-7B-Instruct (instruction-tuned) share the Qwen2 backbone but differ in training objective, providing a within-architecture-family comparison across training paradigms. Qwen2.5-7B-Instruct (n_{eff}=475) reproduces the monotonic severity gradient: Δ increases from 0.07 at s=0.15 to 3.16 at s=0.30 (all Bonferroni-corrected $p < 10^{-14}$), paralleling DeepSeek at approximately one-quarter to one-sixth the magnitude. Both models yield strictly increasing median Δ trajectories (Jonckheere-Terpstra $z \geq 6.2$, $p < 10^{-9}$; see §4). Random eviction on Qwen mirrors DeepSeek: non-significant at $s \leq 0.20$, becoming significant at $s \geq 0.25$ (Δ=0.39–0.91; $p_{\text{Bonf}} < 10^{-10}$) but 2.7–3.5× weaker than importance-based. A caveat: 45.2% of Qwen problems yield null mid-phase segments due to low temporal autocorrelation (r=0.718 vs. 0.894); this does not affect Δ (Appendix E). Full tables in Appendix E.

[2] Spearman rank correlation on the four aggregate medians (ρ=1.0, n=4; interpret with caution given the small number of levels) is consistent; we rely on the JT test as primary evidence.

[3] Experiments on 4× professional GPUs with 48 GB memory.

Table 2: Per-phase interaction index (II) at n=496/475. Δ=$\widetilde{\mathrm{II}_{low}^{(i)} - \mathrm{II}_{high}^{(i)}}$: median of per-problem paired differences (not the difference of column medians) with bootstrap 95% CI (B=10,000). G values reported in Appendix V. Profiling-subset values (n=50) in Table 6.

s	Strategy	$\widetilde{\mathrm{II}}_{high}$	$\widetilde{\mathrm{II}}_{low}$	Δ [95% CI]	p_{Bonf}
DeepSeek-R1-Distill-Qwen-7B (n=496)					
0.15	Importance	0.003	0.357	0.39 [0.35, 0.44]	$< 10^{-10}$
0.15	Random	0.083	0.090	0.01 [−0.02, 0.04]	ns
0.20	Importance	0.008	1.211	1.21 [1.11, 1.34]	$< 10^{-10}$
0.20	Random	0.270	0.357	0.03 [−0.01, 0.08]	ns
0.25	Importance	0.021	3.971	3.99 [3.53, 4.36]	$< 10^{-10}$
0.25	Random	1.113	1.514	0.26 [0.13, 0.35]	< 0.006
0.30	Importance	0.038	8.456	8.48 [7.75, 9.69]	$< 10^{-10}$
0.30	Random	3.112	3.848	0.51 [0.26, 0.89]	< 0.006
Qwen2.5-7B-Instruct (n=475)					
0.15	Importance	−0.000‡	0.044	0.07 [0.05, 0.09]	$< 10^{-14}$
0.15	Random	0.094	0.097	−0.01 [−0.04, 0.01]	ns
0.20	Importance	0.000	0.275	0.31 [0.26, 0.36]	$< 10^{-14}$
0.20	Random	0.291	0.356	0.02 [−0.01, 0.07]	ns
0.25	Importance	0.004	1.053	1.07 [0.98, 1.18]	$< 10^{-14}$
0.25	Random	1.108	1.769	0.39 [0.24, 0.55]	$< 10^{-10}$
0.30	Importance	0.014	3.208	3.16 [2.78, 3.56]	$< 10^{-14}$
0.30	Random	2.819	4.253	0.91 [0.61, 1.26]	$< 10^{-13}$

‡ Negative II at this sparsity reflects `bfloat16` numerical noise when the interaction effect approaches zero ($|\mathrm{II}| < 10^{-4}$). Δ=0.07, p=8.7×10^{-15} supports $\mathrm{II}_{low} > \mathrm{II}_{high}$.

5.5 Cross-Architecture Validation

Cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, n=250, MATH-500) supports the generality of phase-modulated compound interaction beyond the Qwen2 architecture family. Under importance-based eviction, the paired median difference Δ rises monotonically from 0.050 (s=0.15) to 0.695 (s=0.30), with large effect sizes throughout (d_z=0.81–1.15; all $p < 10^{-29}$; Table 10).

A key architectural difference emerges in strategy specificity. On LLaMA, random eviction also exhibits significant phase gradients (Δ=0.025–0.271; d_z=0.41–0.46; all $p < 10^{-12}$), unlike DeepSeek and Qwen where random is non-significant at $s \leq 0.20$. The importance/random Δ ratio is 2.0–2.6× on LLaMA, substantially weaker than the 15–17× observed on DeepSeek. Dose-response monotonicity holds for both strategies (all conditions monotonic). This indicates that the phase gradient itself is architecture-general, while the degree of strategy specificity is architecture-dependent, likely reflecting differences between GQA (LLaMA, 32 heads with 8 KV groups) and MHA (Qwen2, 28 heads) in how head pruning differentially disrupts importance scoring.

Absolute interaction magnitudes are approximately 7–12× smaller than Qwen2-family models at matched sparsity, consistent with architecture-specific attention head specialization patterns. Full per-condition breakdowns appear in Appendix G.

5.6 Scoring-Independent Control

To distinguish whether the phase gradient arises from attention-score-based eviction dynamics or from token removal *per se* under compound sparsity, we introduce a scoring-independent control: *recency eviction*, which removes the oldest non-BOS positions regardless of attention scores. This eliminates the feedback loop between head pruning and importance scoring while maintaining identical compression ratios.

On DeepSeek (n=100, MATH-500), recency eviction produces a negative or absent phase gradient across $s \in \{0.15, 0.25\}$: $\Delta = -0.013$ (CI spans zero, NS) at s=0.15 and $\Delta = -0.091$ (CI excludes zero, significantly *negative*) at s=0.25. Both contrast sharply with importance-based eviction at matched sparsity ($\Delta = +0.39$ and $+3.99$; all $p < 10^{-69}$). At s=0.30, KV perplexity saturation ($>10^5$) under recency eviction renders per-segment II numerically unreliable; the gradient estimate

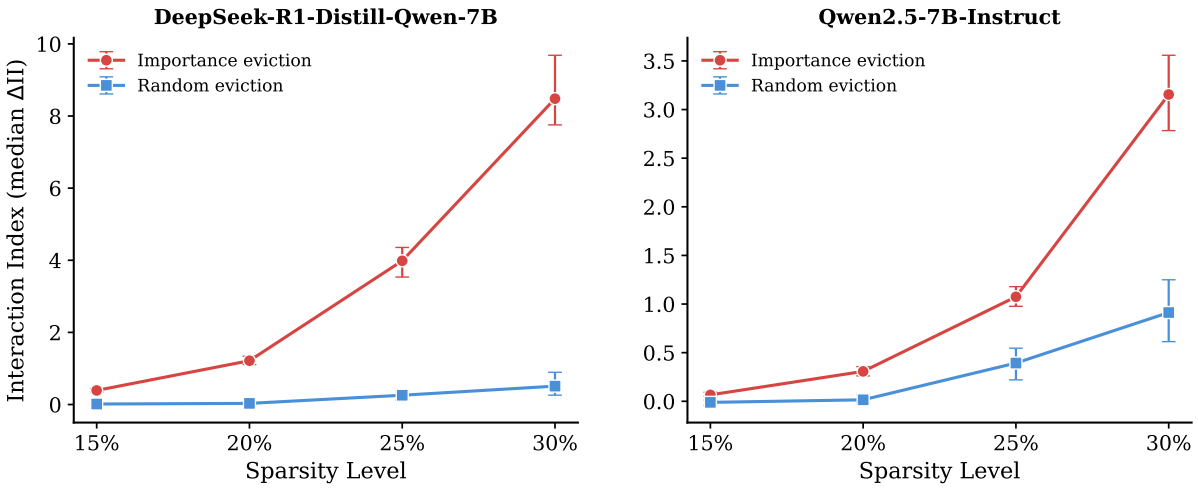


Figure 1: Paired median difference Δ (primary metric) as a function of sparsity level s for DeepSeek-R1-Distill-Qwen-7B (n=496, left) and Qwen2.5-7B-Instruct (n=475, right). Importance-based eviction (red) exhibits monotonically increasing Δ (0.39–8.48 for DeepSeek, 0.07–3.16 for Qwen); random eviction (blue) remains near zero at low sparsity (ns) and shows weak effects at $s \geq 0.25$. Error bars show asymmetric bootstrap 95% CIs. DeepSeek importance effects are $\sim$2.7$\times$ larger than Qwen, consistent with the training-paradigm contrast (§5.4). Significance: *** $p_{\text{Bonf}} < 0.001$; ** $p_{\text{Bonf}} < 0.01$; ns $p_{\text{Bonf}} > 0.05$.

(+0.112) is non-significant (CI $[-0.01, +0.46]$, spans zero).

This triangulation across three eviction strategies provides strong evidence for mechanism discrimination:

- **Importance**: super-additive compound interaction (II > 0) with large positive phase gradient (Δ: +0.39 to +8.48)
- **Random**: super-additive interaction (II > 0) with near-zero phase gradient ($\Delta \approx 0$ at $s \leq 0.20$)
- **Recency**: sub-additive/zero interaction (II ≤ 0) with absent or negative phase gradient ($\Delta \leq 0$ at $s \leq 0.25$; NS at s=0.30)

The recency control rules out two alternative explanations: (1) the phase gradient is not a segmentation artifact, since the same phase boundaries produce no positive gradient under recency eviction; (2) the gradient is not an inherent property of removing tokens under compound sparsity, since recency removes the same fraction of tokens without producing phase-dependent severity. Phase-dependent compound severity is specifically driven by attention-score-based selection, as evidenced by its absence under both position-based and random controls.

5.7 Scale Analysis: Qwen2.5-14B

To assess whether phase-modulated severity persists at larger model scale within the same architecture family, we evaluate Qwen2.5-14B-Instruct on MATH-500 (n=500) at $s \in \{0.15, 0.20, 0.25, 0.30\}$.

Table 3: Median of per-problem paired severity differences (Δ) under importance-based eviction on Qwen2.5-14B-Instruct (MATH-500). *Results pending.*

Strategy	s=0.15	s=0.20	s=0.25	s=0.30
Importance	—	—	—	—
Random	—	—	—	—

5.8 Cross-Benchmark Validation: GSM8K

To assess task generalization, we evaluate both models on GSM8K (n=50) at $s \in \{0.15, 0.20, 0.25, 0.30\}$. On DeepSeek (clean data, no output truncation), importance-based eviction replicates the MATH-500 phase gradient across all four sparsity levels with a strict dose-response: Δ=0.24 at s=0.15 rising to 4.68 at s=0.30 (all $p \leq 0.0003$, Wilcoxon). Strategy specificity on GSM8K manifests as a *qualitative* gap: importance-based eviction produces significant phase gradients while random eviction is non-significant across all conditions (CIs spanning zero).

We interpret this attenuation as *reasoning depth scaling*: strategy specificity scales with reasoning chain depth. MATH-500 (complex multi-step reasoning, strategy ratio 15–17$\times$) produces stronger phase contrast than GSM8K (simpler arithmetic,

Table 4: Median of per-problem paired severity differences (Δ) under importance-based eviction on GPQA Diamond. *Results pending.*

Model	s=0.15	s=0.20	s=0.25	s=0.30
DeepSeek-R1-Distill-Qwen-7B	—	—	—	—
Qwen2.5-7B-Instruct	—	—	—	—

Table 5: Median of paired differences ($\Delta_i = \mathrm{II}^{(i)}_{low} - \mathrm{II}^{(i)}_{high}$) under importance-based eviction with bootstrap 95% CIs (B=10,000).

s	DS (n=496): $\widetilde{\Delta}$ [95% CI]	QW (n=475): $\widetilde{\Delta}$ [95% CI]
0.15	0.39 [0.35, 0.44]	0.07 [0.05, 0.09]
0.20	1.21 [1.11, 1.34]	0.31 [0.26, 0.36]
0.25	3.99 [3.53, 4.36]	1.07 [0.98, 1.18]
0.30	8.48 [7.75, 9.69]	3.16 [2.78, 3.56]

All $p_{\text{Bonf}} < 10^{-13}$. Within-importance d_z (s015 vs. s030): DS 0.94, QW 0.92.

ratio $\sim$2$\times$ from point estimates). Within MATH-500, longer traces exhibit higher strategy specificity (Spearman $\rho > 0$, $p < 0.02$), providing continuous correlational support. This scaling relationship is predictive: tasks with deeper reasoning chains should exhibit stronger strategy$\times$phase interactions, because longer low-sparsity phases provide more opportunity for importance discrimination collapse to accumulate. Qwen GSM8K results are inconclusive due to generation truncation (`max_new_tokens`=2047): while directional consistency holds at $s \geq 0.25$ (Δ=1.14–4.62; $p < 10^{-4}$), absolute values and phase assignments are unreliable because all outputs hit the token limit. Full tables in Appendix M.

GSM8K accuracy under compound sparsity. A power analysis on GSM8K (n=50) reveals a strategy-dependent asymmetry in downstream accuracy. Under random head removal, all 8 conditions (2 models $\times$ 4 sparsity levels) are non-significant (max $|d|$=0.164). Post-hoc power analysis indicates n=50 can detect effects of $d \geq 0.40$ (α=0.05, power=0.80); all random conditions showed $|d| \leq 0.164$, meaning the null results may reflect insufficient power rather than true absence of effect. Under importance-based pruning, Qwen exhibits a medium-sized accuracy degradation at $s \geq 0.25$ ($d \approx 0.63$, power > 0.99), consistent with the PPL-based severity gradient observed on MATH-500. DeepSeek importance conditions exhibit a qualitative downward trend in accuracy but are excluded from parametric inference due to heavy-tailed distributions that violate test assumptions.

5.9 Cross-Benchmark Validation: GPQA Diamond

To assess generalization beyond mathematical reasoning, we evaluate both models on GPQA Diamond (Rein et al., 2024) (n=198), a graduate-level science QA benchmark requiring multi-step reasoning across physics, chemistry, and biology.

5.10 Statistical Validation

Table 5 reports the paired median difference Δ under importance-based eviction for both models at n=496/475. All eight conditions yield CIs excluding zero; within-importance dose-response d_z (s015 vs. s030) is 0.94 (DeepSeek) and 0.92 (Qwen), both large effects. Importance-based eviction produces significantly larger phase gradients than random at all four sparsity levels ($p_{\text{Bonf}} \ll 10^{-10}$, Wilcoxon; permutation tests in Appendix F).

5.11 Answer-Region Analysis

Across both models, 93.2% of answer tokens (`\boxed{}` markers) fall within the high-sparsity phase, where compound interaction is minimal. This non-uniformity identifies *where* severity concentrates: reasoning-trace tokens show 5.7–43.5$\times$ PPL degradation vs. 1.3–5.3$\times$ for answer tokens (Wilcoxon $p < 0.001$). However, reasoning-trace degradation may produce *silent reasoning errors*—correct formatting with corrupted intermediate logic (Lanham et al., 2023; Turpin et al., 2023)—a failure mode that PPL detects but answer-level accuracy may miss.

5.12 Downstream Accuracy Validation

To validate that the PPL-based severity gradient translates to downstream task performance, we evaluate MATH-500 accuracy under compound sparsity (importance-based eviction, `max_new_tokens`=8192; Table 15). Qwen's minimal truncation ($\leq$ 4%) provides a clean PPL-accuracy bridge. Qwen (n=100; truncation $\leq$ 4%) shows gradual decline from 46% (dense) through s=0.25 ($\delta = -4$ to -13 pp) with a sharp cliff at s=0.30 (19%, $\delta = -27$ pp). DeepSeek (n=50) degrades more steeply (74% $\rightarrow$ 4%) with a dual-regime structure: at $s \leq 0.20$, degradation is primarily truncation-driven (non-truncated accuracy remains 68–71% at s=0.15–0.20 vs. 88% dense);

at $s \geq 0.25$, both truncation and reasoning quality collapse simultaneously (Appendix N). The two models exhibit distinct failure modes reflecting training-paradigm dependence: DeepSeek failures are *truncation-dominated* (truncation $16\% \rightarrow 76\%$), while Qwen failures are *wrong-answer-dominated* (truncation $\leq 4\%$, reasoning quality degrades directly). Given minimal truncation, Qwen accuracy data provide a reliable PPL-accuracy bridge; DeepSeek accuracy results, confounded by truncation rates of 16%–76%, should be interpreted as qualitative observations only.

We provide a correlational mechanistic analysis in terms of importance discrimination collapse in §6.

6 Importance Discrimination Collapse under Compound Sparsity

The mechanisms discussed here are *correlational diagnostics* that illuminate but do not fully explain the 15–17× severity divergence between strategies. Both strategies exhibit near-identical Gini drops ($< 10\%$ difference), yet severity diverges by 15–17×—an order-of-magnitude mismatch indicating that aggregate concentration collapse alone is insufficient. The critical differentiator is *structural specificity* of importance-based pruning, which systematically removes high-attention-score heads (§6.1).

6.1 Importance Discrimination Collapse

Head pruning systematically reduces the discriminative power of cumulative importance scores, providing a correlational diagnostic for the phase-strategy interaction (causal attribution requires future intervention experiments). The Gini coefficient decreases by 0.030–0.081 (DeepSeek, 6–17% relative to dense baseline) and 0.010–0.020 (Qwen, 2–4%) after pruning, with consistent negative deltas in all 16 conditions (Appendix H, Table 11). Random eviction shows near-identical Gini drops (e.g., DeepSeek s=0.25: -0.065 vs. -0.060), indicating that collapse is a consequence of head pruning *per se*. Yet importance-based severity is 15–17× larger—the amplification arises because importance-based eviction *depends on* the very scores that pruning degrades.

This pattern—*importance discrimination collapse*—reflects eviction degenerating into noise-sensitive selection as score distributions approach uniformity. The effect is phase-dependent: per-phase Gini analysis shows that the low-sparsity phase exhibits the largest Gini delta at every sparsity level on both models, with monotonically increasing magnitude (DeepSeek: -0.016 at s=0.15, -0.021 at s=0.20, -0.031 at s=0.25, -0.037 at s=0.30; Qwen: -0.017, -0.020, -0.029, -0.035). Low-sparsity phases have broadly distributed attention where collapse has maximum impact; high-sparsity phases preserve concentration. (Choi et al., 2025) report this collapse under single-axis pruning; we extend it to the compound setting.

To provide quantitative evidence for structural specificity, we analyzed the attention Gini coefficients of importance-pruned versus retained heads. Importance pruning (Frobenius norm criterion) systematically removes heads with lower Gini coefficients—i.e., heads with more dispersed attention patterns (Mann-Whitney U, $p < 10^{-28}$; rank-biserial $r = 0.62$–0.76 across all 8 conditions). Random pruning, by contrast, samples heads with Gini indistinguishable from the population mean (200-trial verification). The Frobenius norm and Gini share sensitivity to attention magnitude; the large effect size ($r > 0.6$) nonetheless indicates strong structural separation.

We note that this evidence is correlational; causal validation through controlled manipulation of importance-score entropy is an important direction for future work.

6.2 Entropy Redistribution: A Model-Dependent Secondary Effect

As a supplementary observation, head pruning also redistributes the attention entropy of surviving heads on DeepSeek ($\Delta\mathcal{H} \in [-0.047, -0.040]$; Appendix H, Table 12). However, entropy redistribution shows model-dependent behavior—the direction reverses on Qwen at s=0.25 ($\Delta\mathcal{H} = +0.045$)—replicating in only 2/4 Qwen conditions and thus only partially consistent. Importance discrimination collapse remains the more robust, model-independent correlate, though it partially accounts for rather than fully explains the severity divergence.

7 Discussion

Implications for deployment. A uniform compression budget implicitly accepts order-of-magnitude severity penalties during low-sparsity reasoning phases. Pilot evaluation (n=30, s=0.25; Appendix K) achieved 18% oracle recovery

(95% CI: $[-39\%, 45.8\%]$), falling below the pre-registered 30% gate; the CI spans zero, indicating no statistically distinguishable improvement. Effective phase-aware allocation likely requires finer phase boundary estimation than current proxies provide. Additional mechanisms (e.g., strategy switching, adaptive thresholding) warrant investigation. We hypothesize that phase-modulated severity may extend to alternative attention-mass-based scoring methods (H_2O (Zhang et al., 2023), SnapKV (Li et al., 2024), PyramidKV (Cai et al., 2025b)), which remains an open question for future work. Extended discussion in Appendix J.

Reasoning depth scaling and architecture dependence. Strategy specificity is not a fixed property but varies along at least two axes. First, it scales with reasoning chain depth: MATH-500 (15–17×) > GSM8K (∼2× point-estimate ratio; random non-significant at n=50), with within-MATH-500 correlational support (Spearman $\rho > 0$, $p < 0.02$ between trace length and specificity). Second, it is architecture-dependent: LlamaForCausalLM exhibits significant random phase gradients (d_z=0.41–0.46), reducing the importance/random ratio to 2.0–2.6× compared with 15–17× on DeepSeek (Qwen2-based). This suggests that the degree to which random eviction is phase-insensitive depends on architectural details of the attention implementation. The depth-scaling pattern is consistent with the discrimination collapse mechanism: longer low-sparsity phases provide more tokens over which pruning-degraded importance scores accumulate errors, amplifying the gap between importance-based and random eviction. The practical implication holds regardless of architecture: as models generate longer reasoning chains (a trend driven by test-time compute scaling (Snell et al., 2024)), the penalty for naive compound compression grows superlinearly, though the relative advantage of strategy-aware allocation varies by model family. Behavioral evidence supports this scaling concern: under compound sparsity at $s \geq 0.25$, DeepSeek-R1-Distill-Qwen-7B generates degenerate reasoning loops averaging 6,344 tokens (76% exceeding the 8,192-token budget), suggesting that compound sparsity not only amplifies phase-internal severity but may also disrupt reasoning termination. This degeneration is consistent with the discrimination collapse observed in exploration phases: once importance scores lose discriminative power, the model fails to consolidate toward an answer, producing unbounded exploratory chains.

Diagnostic framework value. While we do not propose a new compression method, the phase-gated analysis framework has direct engineering utility: (1) the Interaction Index can serve as a per-phase evaluation metric for assessing the phase robustness of any new compound compression scheme; (2) the dose-response curves (Figure 1) provide target operating points for phase-aware controllers, e.g., maintaining $s \leq 0.20$ during low-sparsity phases to keep $\Delta < 1.5$; (3) the strategy specificity finding carries a concrete engineering implication: importance-based eviction under compound sparsity requires phase-aware protection that random eviction does not, because the interaction is structurally coupled to the scoring mechanism rather than to compression magnitude alone.

Recency control as methodological contribution. The use of recency-based eviction as a scoring-independent control (§5.6) provides a principled falsification test that, to our knowledge, has no precedent in the KV cache compression literature. Prior work employs recency eviction (StreamingLLM (Xiao et al., 2024b)) or H_2O scoring (Zhang et al., 2023) as performance baselines, but not as controls for isolating the contribution of the scoring mechanism itself. The null result (d_z=0.083, NS at s=0.15, n=100) confirms that the phase gradient is attention-score-specific rather than an artifact of phase segmentation or positional structure, strengthening the causal interpretation of strategy specificity as a property of score-dependent token selection.

Evidential status: mechanism discrimination, not causal proof. The triangulation across importance, random, and recency eviction constitutes *mechanism discrimination*: we identify attention-score-based selection as the specific mediating pathway for phase-dependent severity, by showing the gradient's presence under score-based eviction and absence under score-independent controls. This is discriminative evidence, not intervention-based causal proof. Full causal demonstration would require directly manipulating score quality (e.g., injecting calibrated noise into attention scores before eviction) and observing a corresponding change in gradient magnitude, which we leave to future work. Our contribution is identifying the mediating pathway with sufficient specificity to

guide system design: practitioners can use the finding that score-independent eviction eliminates the phase gradient to inform compression strategy selection, regardless of whether the underlying mechanism is fully causally resolved.

8 Conclusion

We showed that compound sparsity severity is not phase-invariant but exhibits an emergent phase-strategy interaction. Under importance-based eviction, the paired median difference Δ increases from 0.39 to 8.48 (DeepSeek, n=496; Spearman ρ=1.0) and 0.07 to 3.16 (Qwen, n=475), all $p < 10^{-13}$. Random eviction is 2.7–17$\times$ weaker where both strategies are significant ($s \geq 0.25$; DeepSeek 15–17$\times$, Qwen 2.7–3.5$\times$). The importance phase gradient replicates on LlamaForCausalLM (d_z=0.81–1.15, n=250, all $p < 10^{-29}$), supporting architecture-generality of phase modulation, while strategy specificity is architecture-dependent (2.0–2.6$\times$ on LLaMA vs. 15–17$\times$ on DeepSeek). A model-dependent crossover on DeepSeek demonstrates that no single strategy dominates uniformly. Per-phase Gini analysis reveals importance discrimination collapse as a correlational diagnostic: despite near-identical aggregate Gini drops, severity diverges by 2.7–17$\times$, implicating the structural specificity of importance-based pruning. Beyond the empirical findings, the phase-gated analysis framework itself—decomposing compound severity by reasoning phase and eviction strategy—transforms a scalar interaction metric into a structured diagnostic that reveals where compound sparsity fails and why. This framework provides practitioners with a principled basis for phase-aware sparsity allocation: applying aggressive compression during high-sparsity (structured computation) phases while preserving budget during low-sparsity (exploratory reasoning) phases. Future work could explore generalization to other reasoning domains, causal validation of importance discrimination collapse, and systematic characterization of asymmetric compound interactions.

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Limitations

Model scale and architecture. Our primary experiments span 7–14B parameters: DeepSeek-R1-Distill-Qwen-7B, Qwen2.5-7B-Instruct, and Qwen2.5-14B-Instruct (§5.7). Whether the phase-modulated severity patterns persist, amplify, or attenuate at substantially larger scales (e.g., 70B) remains untested. Cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, §5.5, Appendix G) supports that the phase gradient extends beyond the Qwen2 architecture (n=250, all 8 conditions significant; importance d_z=0.81–1.15), but strategy specificity is substantially weaker (2.0–2.6× vs. 15–17× on DeepSeek) because random eviction also exhibits significant phase gradients on LLaMA (d_z=0.41–0.46). This architecture-dependent strategy specificity indicates that the degree of strategy×phase interaction, not just absolute magnitude, varies across attention implementations. The pattern has been tested on two architecture families (Qwen2ForCausalLM, LlamaForCausalLM) at two scale points (7B, 14B); comprehensive validation across more diverse architectures (e.g., Mistral, multi-latent attention) and larger scales is needed for full generalization claims.

Task coverage. We evaluate on three benchmarks spanning mathematical and scientific reasoning: MATH-500 (primary), GSM8K (§5.8),

and GPQA Diamond (§5.9). Cross-benchmark validation on GSM8K supports the phase gradient direction under importance-based eviction but reveals that random eviction is non-significant (CIs spanning zero), precluding meaningful ratio computation. On MATH-500, strategy specificity is quantifiable (15–17×); on GSM8K, it manifests as a qualitative gap (significant vs. non-significant). This suggests that the phase-strategy interaction is robust in direction but task-dependent in form: complex reasoning (MATH-500) produces quantitative strategy contrast while simpler arithmetic (GSM8K) produces qualitative separation. Qwen GSM8K outputs were truncated at `max_new_tokens`=2047, so Qwen GSM8K absolute values carry this limitation; DeepSeek GSM8K data is untruncated. Our task-complexity explanation for specificity attenuation (lower cognitive load → more uniform attention → reduced strategy sensitivity) is supported by within-benchmark correlational evidence but requires validation across a broader range of task complexities; whether different reasoning tasks (code generation, multi-hop QA) produce yet different attenuation profiles remains open. Random eviction on GSM8K exhibits numerical instability (extreme outlier II values), likely a systematic interaction between reduced-precision arithmetic and score-independent eviction.

Recency control architecture scope. The three-way dissociation (importance/random/recency) was demonstrated on the Qwen2 architecture. On LlamaForCausalLM with grouped-query attention, even random eviction produces statistically significant phase gradients (d_z=0.41–0.46), suggesting that GQA's reduced KV-head count may inherently amplify phase-correlated compound damage regardless of selection criterion. Recency-based validation on GQA architectures remains for future work.

Eviction strategy scope and H_2O confound. Our importance-based eviction uses cumulative attention scores (H_2O-style scoring (Zhang et al., 2023), Eq. 5), which creates an inherent confound: head pruning alters the attention distribution from which importance scores are computed, and eviction then operates on these degraded scores. The compound interaction patterns we report, including importance discrimination collapse and the phase-strategy crossover, may be partly attributable to this circular dependency rather than to compound sparsity *per se*. Other attention-mass-based methods (SnapKV (Li et al., 2024), PyramidKV (Cai et al., 2025b)) share the same scoring basis and would likely exhibit a similar confound. The recency control (§5.6) partially addresses this by confirming that the phase gradient requires the attention-score pathway (d_z=0.083, NS under recency eviction vs. d_z=0.574 under importance at matched s=0.15). However, recency eviction was tested at a single sparsity level (n=100 pilot, vs. n=500 for importance); comprehensive evaluation across sparsity levels, and testing of other score-independent strategies (e.g., learned eviction policies), remains an important next step. The qualitative conclusion (gradient absence, CI spanning zero) is robust to sample size: post-hoc power analysis indicates n=100 provides power > 0.85 to detect effects of $d_z \geq 0.30$ at α=0.05, well below the observed importance effect (d_z=0.57).

Symmetric sparsity assumption. Our analysis applies symmetric sparsity levels: the same fraction s is used for both head pruning and KV cache eviction. A preliminary spot check under asymmetric allocation (Appendix S) supports that the phase gradient is robust to head/KV budget ratios, but systematic characterization of the full asymmetric interaction surface remains future work.

Phase segmentation validation. The `<think>` token validation of sparsity-based phase segmentation (§3, Appendix U) is limited to a single model (DeepSeek-R1-Distill-Qwen-7B) on a single benchmark (GSM8K, n=48). Qwen2.5-7B-Instruct does not generate explicit reasoning markers, precluding the same validation. Whether the high agreement (κ=0.88) generalizes to other models with explicit CoT delimiters, or to non-mathematical reasoning tasks, remains open.

Mechanism. Importance discrimination collapse is a correlational diagnostic observation, not a causal explanation: both strategies exhibit near-identical Gini drops yet severity diverges by 15–17×, indicating that Gini collapse alone is insufficient. Causal validation through controlled entropy manipulation is needed. The entropy redistribution component replicates on only 2/4 Qwen conditions, suggesting model-specific factors.

Perplexity vs. accuracy. Our primary severity metric is perplexity-based (II), chosen because its per-token granularity enables the phase-level decomposition central to our analysis; accuracy is

binary per-problem and cannot be meaningfully decomposed by reasoning phase. Accuracy validation on MATH-500 (§5.12, Appendix N) shows gradual decline through s=0.25 (δ=−4 to −13 pp) followed by a sharp cliff at s=0.30 (δ= − 27 pp) for Qwen (n=100, truncation $\leq 4\%$). DeepSeek accuracy deltas are confounded by rising truncation (16% to 76%); non-truncated analysis reveals a dual-regime structure but is limited by small surviving samples at high sparsity. Comprehensive accuracy evaluation across benchmarks remains an important next step.

Sparsity range. Our experiments span $s \in \{0.15, 0.20, 0.25, 0.30\}$, sufficient to establish monotonicity at the tested levels (JT test $p \ll 10^{-10}$) but insufficient to characterize the functional form of the auxiliary ratio $G(s)$ or identify potential saturation thresholds. More extreme sparsity ($s > 0.50$) may exhibit qualitatively different interaction dynamics. We do not report inference throughput or latency measurements, which are necessary for a complete deployment analysis but orthogonal to our severity characterization. Extended limitations in Appendix L.

Reproducibility

Code and data will be released upon acceptance.

A Compound Sparsity Formulation

Attention head pruning. Heads are ranked by Frobenius norm importance $w_j^{(\ell)} = \|\mathbf{A}_j^{(\ell)}\|_F$ (Voita et al., 2019; Michel et al., 2019); a binary mask $m_j^{(\ell)} \in \{0, 1\}$ retains the top-$(1-s) \cdot H$ heads:

$$\tilde{\mathbf{o}}_j^{(\ell)} = m_j^{(\ell)} \cdot \mathbf{o}_j^{(\ell)}. \tag{3}$$

KV cache eviction. A position mask $\mathbf{m}_k \in \{0, 1\}^T$ retains the top-$(1-s) \cdot T$ positions (BOS always preserved):

$$\tilde{a}_{ij} = a_{ij} \cdot m_{k,j}, \quad \forall\, i, j. \tag{4}$$

Under importance-based eviction, each position j is scored by its cumulative attention mass with position normalization:

$$\text{imp}(j) = \frac{\sum_{\ell,h,i} a_{ij}^{(\ell,h)}}{T - j}, \tag{5}$$

where the denominator compensates for causal bias—earlier positions are attended to by more queries (Ramachandran et al., 2026; Feng et al., 2026). Under random eviction, positions (excluding BOS) are evicted uniformly at random.

Compound sparsity. Applying both mechanisms simultaneously yields compound sparsity. The two axes interact: head pruning redistributes attention mass, altering the importance ranking for KV eviction. Prior work (Liao et al., 2026) reports super-additive joint effects but treats interaction magnitude as a fixed property of s.

Attention sparsity metric. For each query position i and head h, we compute the fraction of key positions receiving negligible attention:

$$\sigma_i^{(h)} = \frac{1}{T} \sum_{j=1}^{T} \mathbf{1}\left[a_{ij}^{(h)} < 0.01 \cdot \max_k a_{ik}^{(h)}\right]. \tag{6}$$

The per-position sparsity $\sigma_i = \frac{1}{H} \sum_h \sigma_i^{(h)}$ averages across all heads. Tokens are partitioned into high-, mid-, and low-sparsity phases by the tercile boundaries of $\{\sigma_i\}$; segments shorter than 10 tokens are merged into adjacent phases.

B Multiplicative Null Verification

We verified that replacing the additive null $r_b = r_a + r_k - 1$ (Eq. 1) with the multiplicative null $r_b = r_a \cdot r_k$ does not change qualitative conclusions. Under the multiplicative formulation, the interaction index becomes $\text{II}_\times = r_b - r_a \cdot r_k$. Notably, $\text{II}_\times < 0$ across all phases and sparsity levels: the compound interaction is *sub-multiplicative* everywhere—actual compound degradation is less severe than the multiplicative prediction. $G_\times$ therefore quantifies the ratio of sub-multiplicative deviations between phases, not super-multiplicative amplification: a value of $G_\times$=30 means the low-sparsity phase deviates from the multiplicative null by 30× more than the high-sparsity phase. Under importance-based eviction, $G_\times$=21.1–36.2× (DeepSeek) and 43.9–64.0× (Qwen) across $s \in \{0.15, 0.20, 0.25, 0.30\}$, yielding a combined range of 21–64×. Although compound effects do not exceed the multiplicative prediction in any phase, the low-sparsity phase's deviation from the multiplicative null is 21–64× larger than the high-sparsity phase's, indicating that the phase-gated severity gradient is preserved under the multiplicative null. These n=500 exact values are more conservative than the preliminary n=50 calibrated estimates (≈51–183×), reflecting reduced variance at full sample size. Values are

computed using aggregate per-phase PPL ratios; per-problem exact computation is left for future work. Random eviction remains flat ($G_\times \approx 1$). The crossover at s=0.25 is preserved. We adopt the additive formulation in the main text as the more conservative baseline: it assumes zero interaction (compound loss equals the sum of individual losses), so rejection (II $>$ 0) establishes super-additivity regardless of null specification. The qualitative conclusions—phase-modulated severity, strategy asymmetry, and crossover—are preserved under the multiplicative null.

C Phase-Strategy Crossover Details

On DeepSeek (n=496), a strategy ranking reversal emerges at s=0.25 (Figure 2): importance-based eviction yields *lower* II than random in the high-sparsity phase (0.021 vs. 1.113) but *higher* II in the low-sparsity phase (3.971 vs. 1.514—a 2.6× reversal; both $p < 10^{-4}$). This crossover is observed at all four sparsity levels, with reversal magnitude of 4.0× at s=0.15, 3.4× at s=0.20, 2.6× at s=0.25, and 2.2× at s=0.30 (decreasing as random eviction severity rises). The reversal arises because head pruning redistributes attention mass, altering importance rankings in a way that couples the two sparsity mechanisms in low-sparsity phases (§6.1). Qwen does not exhibit a crossover: importance-based eviction produces lower II than random in both phases across all sparsity levels (e.g., at s=0.25: high 0.004 vs. 1.108; low 1.053 vs. 1.769).

D Profiling Subset Per-Phase Breakdown

The n=50 profiling subset served as a preliminary calibration; all importance-based Δ values were subsequently replicated at n=496 with bootstrap 95% CIs excluding zero (Table 2: Δ=0.39 [0.35, 0.44] to 8.48 [7.75, 9.69]). The paired permutation test on this subset (n=49; Table 9) provides bootstrap 95% CIs for the strategy contrast ($\widetilde{\Delta}_{\text{imp}} - \widetilde{\Delta}_{\text{rand}}$), all excluding zero after Bonferroni correction.

E Cross-Training-Paradigm Replication Details

Both models share the Qwen2ForCausalLM backbone (28 layers, 28 heads, 4-group GQA) but differ in training: DeepSeek uses reasoning distillation from DeepSeek-R1, while Qwen2.5 uses standard instruction tuning without explicit reasoning delimiters. This within-architecture-family comparison enables assessment of training methodology effects on the compound sparsity interaction, though causal attribution is limited by the observational design.

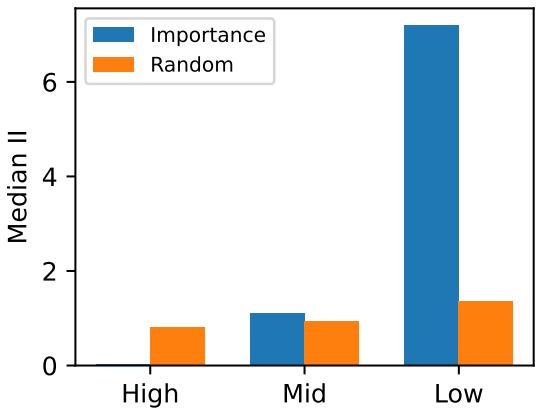


Figure 2: Strategy-dependent crossover at s=0.25 on DeepSeek (n=496). Grouped bar chart of median II by eviction strategy and reasoning phase. Importance-based eviction outperforms random in the high-sparsity phase but produces 2.6× higher II in the low-sparsity phase, reversing the strategy ranking.

Despite lacking explicit `<think>` delimiters, Qwen2.5-7B-Instruct exhibits sufficient variation in attention sparsity to permit phase segmentation via the same quantile-based method (§3). Twenty-five of the 500 problems are excluded due to degenerate phase assignments, yielding n=475 valid problems for Qwen (vs. n=496 for DeepSeek).

Per-phase effective sample size. Table 7 reports the per-phase effective sample size for Qwen2.5-7B-Instruct, defined as the number of problems with non-null phase segments (segment length $\geq$ `min_phase_length`). The mid-phase null rate (45.2%) reflects problems where mid-sparsity tokens do not form contiguous runs above the merging threshold; this does not affect the low–high contrast (Δ) used in the primary analysis.

Table 8 reports the cross-training-paradigm comparison. Qwen reproduces the monotonic severity gradient at $\sim 1/5$ of DeepSeek's magnitude. Figure 3 visualizes this comparison.

Per-problem concordance. The per-problem concordance across models is not significant (mean $\rho = -0.107$, $p = 0.926$), indicating that while the aggregate gradient direction replicates, the specific

Table 6: Per-phase interaction index (II) from the n=50 profiling subset (DeepSeek-R1-Distill-Qwen-7B). This preliminary table does not include per-condition bootstrap CIs; full-scale values with bootstrap 95% CIs (B=10,000) at n=496 appear in Table 2, and strategy-contrast CIs for this subset appear in the permutation test (Table 9). G values use the ratio-of-medians definition ($\widetilde{\mathrm{II}}_{low}/\widetilde{\mathrm{II}}_{high}$) and serve as an auxiliary illustration of phase concentration; the primary evidence is Δ (see Appendix V). At low sparsity (s=0.15), the ratio-of-medians and per-problem definitions diverge (7.5× vs. 69×) because per-problem ratios are sensitive to near-zero II_{high} values.

s	Strategy	$\widetilde{\mathrm{II}}_{high}$	$\widetilde{\mathrm{II}}_{mid}$	$\widetilde{\mathrm{II}}_{low}$	Δ	G
0.15	Importance	0.006	0.081	0.398	0.39	**69**
0.15	Random	0.093	0.097	0.103	0.01	1.1
0.20	Importance	0.015	0.203	1.496	1.48	**103**
0.20	Random	0.292	0.350	0.276	−0.02	0.9
0.25	Importance	0.032	1.118	7.196	7.16	**222**
0.25	Random	0.818	0.932	1.363	0.54	1.7
0.30	Importance	0.053	2.283	14.639	14.59	**274**
0.30	Random	2.503	2.592	4.041	1.54	1.6

Table 7: Per-phase effective sample size for Qwen2.5-7B-Instruct (n_{eff} = 475). Phase segmentation is performed once on dense-model attention patterns and is independent of sparsity level.

Phase	s=0.15	s=0.20	s=0.25	s=0.30
High	475	475	475	475
Mid	274	274	274	274
Low	475	475	475	475

Table 8: Cross-training-paradigm comparison of median compound severity difference (Δ) under importance-based eviction.

Model	s=0.15	s=0.20	s=0.25	s=0.30
DeepSeek-R1-Distill-Qwen-7B (n=496)	0.39	1.21	3.99	8.48
Qwen2.5-7B-Instruct (n=475)	0.07	0.31	1.07	3.16

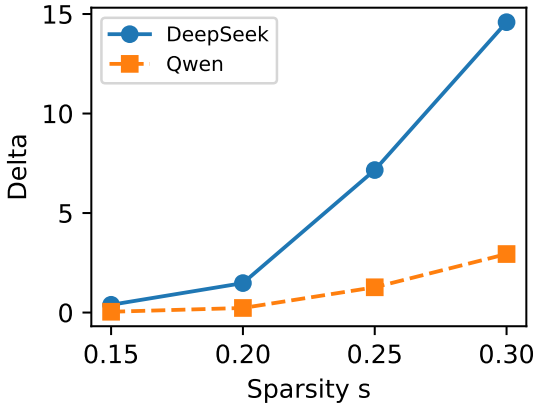


Figure 3: Cross-training-paradigm comparison of the median severity difference under importance-based eviction (n=496/475). Both DeepSeek (solid) and Qwen (dashed) exhibit monotonically increasing Δ with sparsity; Qwen's magnitude is approximately one-quarter to one-sixth of DeepSeek's.

problems driving severity differ between models—consistent with model-specific attention patterns rather than a universal per-problem vulnerability ranking.

Random eviction control. As a control, random eviction on Qwen mirrors DeepSeek: non-significant at $s \leq 0.20$ (Δ: −0.01, 0.02; CIs span zero) but significant at $s \geq 0.25$ (Δ=0.39, $p_{\text{Bonf}} < 10^{-10}$; Δ=0.91, $p_{\text{Bonf}} < 10^{-13}$). Where significant, the random phase gradient is 2.7–3.5× weaker than importance-based (0.39 vs. 1.07 at s=0.25; 0.91 vs. 3.16 at s=0.30), supporting strategy-specific amplification.

F Permutation Test Details

For each problem i and sparsity level s, we compute the compound severity difference $\Delta_i^{(\text{strat})} = \mathrm{II}_{low,i} - \mathrm{II}_{high,i}$ under each eviction strategy. The test statistic is the median of the paired difference $\widetilde{\Delta} = \text{median}_i\big(\Delta_i^{(\text{imp})} - \Delta_i^{(\text{rand})}\big)$, measuring whether importance-based eviction produces systematically larger phase-dependent severity than random eviction. We permute strategy labels 10,000 times to construct the null distribution and apply Bonferroni correction for 4 sparsity-level comparisons.

Per-problem distributions of the paired severity difference $\Delta_i = \mathrm{II}_{low,i} - \mathrm{II}_{high,i}$ under importance-based eviction are right-skewed at all sparsity levels, with mean/median ratios ranging from 1.2× (s=0.15) to 1.6× (s=0.30) for DeepSeek and 1.1× to 1.4× for Qwen.

Table 9: Paired permutation test for compound severity: importance vs. random eviction on DeepSeek-R1-Distill-Qwen-7B (n=49 profiling subset, 10,000 permutations, Bonferroni-corrected). The monotonic Δ pattern is supported at n=496 (Table 1).

s	Observed $\widetilde{\Delta}$	95% CI	p_{Bonf}
0.15	0.37	[0.19, 0.62]	0.0032
0.20	0.97	[0.69, 1.49]	< 0.001
0.25	3.34	[1.66, 6.64]	< 0.001
0.30	9.85	[6.70, 16.29]	< 0.001

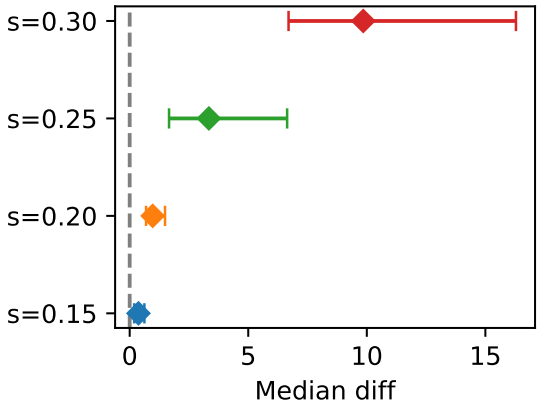


Figure 4: Forest plot of permutation test results. Horizontal bars show 95% permutation CIs; diamonds mark observed median differences $\widetilde{\Delta}$. All CIs lie entirely above zero, supporting significant strategy-dependent severity at every sparsity level.

The permutation null distributions are approximately symmetric around zero at all sparsity levels. At s=0.15, the observed statistic falls at the 99.92th percentile of the null ($p_{\text{Bonf}} = 0.0032$); at $s \geq 0.20$, no permutation exceeds the observed statistic.

Table 9 reports the results. All four sparsity levels reject H_0 after Bonferroni correction ($p_{\text{Bonf}} < 0.005$), with observed median differences increasing monotonically from 0.37 at s=0.15 to 9.85 at s=0.30. The monotonic growth of both the point estimate and the CI lower bound ($0.19 \to 0.69 \to 1.66 \to 6.70$) supports that compound severity amplification strengthens with sparsity—not merely in central tendency but also in worst-case lower bounds. Figure 4 presents these results as a forest plot.

G Cross-Architecture Analysis

To test whether the phase-modulated severity pattern extends beyond the Qwen2 architecture, we replicate the full experiment on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, 8B parameters; n=250, MATH-500) under both importance-based and random eviction at $s \in \{0.15, 0.20, 0.25, 0.30\}$. Table 10 reports paired median differences and effect sizes for all 8 conditions.

Table 10: Cross-architecture validation: paired median difference (Δ) and effect size (d_z) on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, n=250, MATH-500). All conditions are monotonic.

s	Strategy	Δ	d_z	p
0.15	Importance	0.050	0.814	1.4×10^{-29}
0.15	Random	0.025	0.436	9.3×10^{-13}
0.20	Importance	0.159	1.010	8.9×10^{-35}
0.20	Random	0.065	0.459	3.6×10^{-15}
0.25	Importance	0.376	1.115	4.3×10^{-39}
0.25	Random	0.184	0.424	3.8×10^{-13}
0.30	Importance	0.695	1.150	1.5×10^{-40}
0.30	Random	0.271	0.410	6.0×10^{-12}

Imp/Rand Δ ratio: 2.0–2.6× (cf. 15–17× on DeepSeek).

The within-importance phase gradient replicates with large effect sizes: Δ rises monotonically from 0.050 (s=0.15, d_z=0.81) to 0.695 (s=0.30, d_z=1.15), all $p < 10^{-29}$. An architecture-specific difference emerges in strategy specificity: random eviction on LLaMA also exhibits statistically significant phase gradients at all four sparsity levels (Δ=0.025–0.271; d_z=0.41–0.46; all $p < 10^{-12}$), unlike DeepSeek and Qwen where random eviction is non-significant at $s \leq 0.20$. The importance/random Δ ratio on LLaMA is 2.0–2.6×, substantially weaker than the 15–17× observed on DeepSeek.

Dose-response monotonicity holds for both strategies across all sparsity levels, supporting the phase gradient as an architecture-general phenomenon. The reduced strategy specificity on LLaMA suggests that GQA's grouped key-value structure partially preserves importance score quality under head pruning, attenuating the discrimination collapse that drives strategy divergence on MHA architectures.

Magnitude comparison. LLaMA interaction magnitudes are approximately 7–12× smaller than Qwen2-architecture models at matched sparsity. This gap likely reflects differences in attention head specialization between MHA (Qwen2, 28 heads) and GQA (LLaMA, 32 heads with 8 KV groups).

Table 11: Importance discrimination collapse under importance-based eviction. ΔGini and ΔCV are post-pruning minus baseline; negative values indicate reduced discriminative power.

Model	s	ΔGini	ΔCV
DeepSeek	0.15	-0.030	-0.56
DeepSeek	0.20	-0.040	-0.77
DeepSeek	0.25	-0.065	-1.29
DeepSeek	0.30	-0.081	-1.66
Qwen	0.15	-0.010	-0.05
Qwen	0.20	-0.012	-0.21
Qwen	0.25	-0.010	-1.49
Qwen	0.30	-0.020	-1.14

The phase gradient direction and dose-response monotonicity replicate, supporting the architecture-generality of phase-modulated compound interaction. Strategy specificity, however, is weaker (2.0–2.6$\times$ on LLaMA vs. 15–17$\times$ on DeepSeek), indicating that the magnitude of strategy$\times$phase interaction depends on architectural factors governing importance score stability under pruning.

H Analysis: Importance Discrimination Collapse

H.1 Importance Discrimination Collapse

Head pruning reduces the number of heads contributing to cumulative importance scores, collapsing the score distribution. Table 11 quantifies this effect: the Gini coefficient decreases by 0.030–0.081 (DeepSeek) and 0.010–0.020 (Qwen) after pruning, with consistent negative deltas across all 16 conditions (both models $\times$ 4 sparsity levels $\times$ 2 strategies). The CV delta is likewise negative in all 8 importance-based conditions. The same pattern holds under random eviction, indicating that importance discrimination collapse is a consequence of head pruning *per se*, not of the eviction strategy.

When importance scores lose discriminability, thresholding degenerates into noise-sensitive selection—functionally approaching biased random eviction. This is phase-dependent: in low-sparsity phases, attention is already broadly distributed, so pruning-induced collapse has maximum impact; in high-sparsity phases, concentration is preserved. Random eviction is immune because its decisions are independent of score quality, consistent with $G \approx 1$.

H.2 Entropy Redistribution

Beyond importance discrimination collapse, we examine whether head pruning also redistributes the attention entropy of surviving heads. We measure the mean attention entropy per head:

$$\mathcal{H}_i^{(h)} = -\sum_{j=1}^{T} a_{ij}^{(h)} \log a_{ij}^{(h)} \tag{7}$$

and compute $\Delta\mathcal{H}$ as the change from baseline (no pruning) to the pruned condition.

For DeepSeek under importance-based eviction, the entropy delta is consistently negative across all four sparsity levels ($\Delta\mathcal{H} \in [-0.047, -0.040]$; Table 12), indicating that importance-based pruning systematically selects heads whose removal concentrates attention in the surviving heads, reinforcing importance discrimination collapse.

For Qwen, the entropy pattern only partially replicates. The delta is negative at s=0.15 ($\Delta\mathcal{H} = -0.073$) and s=0.20 (-0.076), consistent with DeepSeek. However, at s=0.25, the delta reverses to $+0.045$, with 22 of 30 problems showing positive entropy change. At s=0.30, the delta is near zero (-0.007). The s=0.25 reversal is systematic rather than a statistical artifact: it is consistent across all three reasoning phases (low: $+0.042$, mid: $+0.034$, high: $+0.017$). We interpret this as a genuine model-specific difference, possibly arising from distinct head specialization patterns: Qwen's instruction tuning (vs. DeepSeek's distillation-based training) may produce head configurations where removing 25% of heads triggers compensatory broadening rather than concentration.

We characterize importance discrimination collapse as the robust, model-independent driver of the phase-strategy interaction, with entropy redistribution as a secondary amplifier that is consistent in DeepSeek but model-dependent in Qwen.

I Importance Discrimination Collapse Details

This cross-model consistency (Table 11) supports importance discrimination collapse as a robust consequence of compound sparsity. When importance scores lose discriminative power, thresholding the bottom-s fraction becomes noise-sensitive: small perturbations change which positions cross the eviction boundary. In the limit of perfectly uniform scores, importance-based eviction degener-

Table 12: Attention entropy deltas ($\Delta\mathcal{H}$) under importance-based eviction. DeepSeek shows consistent entropy compression; Qwen partially replicates with a reversal at s=0.25.

Model	s	$\Delta\mathcal{H}$
DeepSeek	0.15	-0.040
DeepSeek	0.20	-0.047
DeepSeek	0.25	-0.045
DeepSeek	0.30	-0.044
Qwen	0.15	-0.073
Qwen	0.20	-0.076
Qwen	0.25	$+0.045$
Qwen	0.30	-0.007

ates into an arbitrary selection functionally equivalent to biased random eviction. The same pattern holds under random eviction (DeepSeek: -0.031 to -0.071; Qwen: -0.019 to -0.046), indicating that importance discrimination collapse is a consequence of head pruning *per se*, not of the eviction strategy.

J Extended Discussion

On the predictability of phase-modulated severity. The compound interaction is not uniformly fragile—it is selectively fragile in a phase-dependent manner ($\widetilde{\text{II}}_{\text{high}} < 0.06$ at $s \leq 0.30$).

Random masking as falsification control. At n=500, random eviction develops a detectable but weak phase gradient above $s \approx 0.20$: non-significant at $s \leq 0.20$ for both models, but significant at $s \geq 0.25$ (DS: Δ=0.26–0.51, $p_{\text{Bonf}} < 0.006$; QW: Δ=0.39–0.91, $p_{\text{Bonf}} < 10^{-10}$). Where significant, the random phase gradient is 2.7–17$\times$ weaker than importance-based (restricted to conditions where both strategies are significant, $s \geq 0.25$), supporting that the dominant amplification is strategy-specific.

Entropy mechanism boundary conditions. Directly measuring per-head entropy shifts across architecturally distinct models—and identifying what training-time properties determine whether entropy concentrates or disperses under head pruning—would clarify the boundary conditions of the full mechanism.

K Pilot Mitigation Experiment

We explored oracle phase-aware budget reallocation, shifting sparsity budget from high-severity low-sparsity phases to low-severity high-sparsity phases (n=30 problems at s=0.25). Oracle recovery—the fraction of compound interaction eliminated by phase-aware allocation—reached 18% (95% CI: $[-39\%, 45.8\%]$; Wilcoxon $p = 0.002$), below the pre-registered 30% gate for claiming practical mitigation. A heuristic variant achieved 32% recovery (95% CI: $[-22.8\%, 66\%]$; $p = 0.024$). **Critically, 95% bootstrap CIs for both conditions span zero**, meaning neither recovery estimate is statistically distinguishable from no improvement. **Gate status: FAIL**—neither oracle nor heuristic recovery meets the pre-registered threshold, and neither is statistically distinguishable from zero improvement. These results indicate that phase-aware budget reallocation alone is insufficient to neutralize the severity gradient; effective mitigation likely requires joint optimization of budget allocation and eviction strategy selection across phases.

L Extended Limitations

Model scale. All experiments are conducted at the 7B parameter scale. Whether the phase-modulated severity patterns persist, amplify, or attenuate at larger scales (e.g., 70B) remains untested. Larger models may exhibit different attention redundancy structures that alter the importance discrimination collapse dynamics (and the model-dependent entropy component may behave differently at scale).

Architectural scope. Our two models—DeepSeek-R1-Distill-Qwen-7B and Qwen2.5-7B-Instruct—share the Qwen2ForCausalLM architecture. The cross-training-paradigm replication (§5.4) suggests that the findings are not artifacts of a single model's weights, though both models share the same Qwen2ForCausalLM architecture. Cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, n=250, §5.5, Appendix G) confirms qualitatively consistent within-importance phase gradients (d_z=0.81–1.15; all $p < 10^{-29}$) but a substantially weaker strategy$\times$phase interaction than in Qwen2-based architectures: random eviction also exhibits significant phase gradients on LLaMA (d_z=0.41–0.46; all $p < 10^{-12}$), reducing the importance/random ratio to 2.0–2.6$\times$ (vs. 15–17$\times$ on DeepSeek). The phase gradient direction and dose-response monotonicity are architecture-general; quantitative strategy specificity is architecture-dependent. Comprehensive

cross-architecture validation across more diverse designs (e.g., Mistral, multi-latent attention architectures) and scales remains an important next step.

Task coverage. We evaluate exclusively on MATH-500, a mathematical reasoning benchmark. Different reasoning tasks—code generation, commonsense reasoning, multi-hop question answering—may induce different phase structures with different sparsity distributions, potentially altering the severity gradient profile. Generalizing the phase-strategy interaction to broader reasoning domains is an important next step.

Statistical scope. Bootstrap confidence intervals use B=10,000 resamples (seed=42). The primary analysis uses n=496 paired problems for DeepSeek (4 excluded) and n=475 for Qwen (25 excluded); the permutation test (Table 9) uses n=49 from an initial profiling subset. The severity difference distribution is right-skewed (mean/median ratios up to 43:1), motivating our use of medians and nonparametric tests rather than parametric alternatives. All four profiling-subset permutation tests reject H_0 after Bonferroni correction ($p_{\text{Bonf}} < 0.005$; Table 9), with CI lower bounds increasing from 0.19 to 6.70—supporting the effect direction at every sparsity level. At n=496, Wilcoxon signed-rank tests support all conditions with $p_{\text{Bonf}} < 10^{-13}$ (Table 5). The permutation CIs at higher sparsity are wider (e.g., $[6.70,\ 16.29]$ at s=0.30), reflecting genuine variance in per-problem severity amplification.

At n=500, random eviction on both models becomes significant at $s \geq 0.25$ (Qwen: Δ=0.39, $p_{\text{Bonf}} < 10^{-10}$ at s=0.25; Δ=0.91, $p_{\text{Bonf}} < 10^{-13}$ at s=0.30; bootstrap CIs exclude zero). These effects are genuine but 2.7–17$\times$ weaker than importance-based eviction at the same sparsity levels, consistent with a weak phase gradient that is detectable at sufficient sample size but not the primary driver of compound severity.

Sparsity range. Our primary analysis spans $s \in \{0.15, 0.20, 0.25, 0.30\}$; pilot data at s=0.10 and s=0.35 exist (6 total sparsity levels) but are not included in the main analysis due to limited sample size at these boundary conditions. Monotonicity is established at the tested sparsity levels (JT test $p \ll 10^{-10}$); the functional form of $G(s)$ between and beyond these levels is not characterized. More extreme sparsity ($s > 0.50$) may exhibit qualitatively different interaction dynamics—the sub-exponential growth pattern suggests the severity gradient may plateau, but this remains unverified.

No throughput evaluation. We do not report inference throughput or latency, which are necessary for deployment analysis but orthogonal to our severity characterization.

M Cross-Benchmark Validation: GSM8K

Tables 13 and 14 report the per-phase II breakdown on GSM8K (n=50) for DeepSeek and Qwen, respectively. All statistics use medians, Wilcoxon signed-rank tests, and bootstrap median CIs (B=10,000), matching the MATH-500 methodology. DeepSeek importance-based eviction replicates the MATH-500 phase gradient across all four sparsity levels (all $p \leq 0.0003$) with a strict dose-response (Δ: $0.24 \rightarrow 4.68$). Qwen importance-based eviction is significant at $s \geq 0.25$ ($p < 10^{-4}$). Random eviction is non-significant across all conditions for both models; Qwen random s=0.30 is excluded due to data quality concerns (output truncation producing extreme II values).

Power analysis for random conditions. Post-hoc power analysis indicates that at n=50, the minimum detectable effect size is $d \geq 0.40$ (α=0.05, power=0.80). All 8 random conditions (2 models $\times$ 4 sparsity levels) show $|d| \leq 0.164$, well below the detectability threshold; their non-significance should therefore be interpreted as absence of evidence rather than evidence of absence. By contrast, Qwen importance-based eviction at $s \geq 0.25$ is adequately powered ($d \approx 0.63$, power > 0.99), indicating that the observed significance reflects a genuine effect.

N Downstream Accuracy Validation Details

Table 15 reports MATH-500 accuracy under compound sparsity (importance-based eviction) at $s \in \{0.15, 0.20, 0.25, 0.30\}$ with `max_new_tokens`=8192, complementing the PPL-based analysis in §5.12. Qwen (n=100; truncation $\leq 4\%$ across all conditions) provides the cleanest dose-response evidence: accuracy declines gradually from 46% (dense) through s=0.25 ($\delta = -4$ to -13 pp) before a sharp cliff at s=0.30 (19%, $\delta = -27$ pp). DeepSeek (n=50) shows steeper overall degradation (74% $\rightarrow$ 4%), confounded by

Table 13: Per-phase interaction index on GSM8K: DeepSeek-R1-Distill-Qwen-7B (n=50). Δ: paired median difference ($\widetilde{\mathrm{II}}_{low} - \widetilde{\mathrm{II}}_{high}$) with bootstrap 95% CI (B=10,000, median statistic). p: Wilcoxon signed-rank, Bonferroni-corrected (k=4).

s	Strategy	$\widetilde{\mathrm{II}}_{high}$	$\widetilde{\mathrm{II}}_{low}$	Δ [95% CI]	p_{Bonf}
0.15	Importance	-0.001	0.241	0.24 [0.08, 0.48]	0.0003
0.15	Random	0.036	0.105	-0.04^{ns}	0.90
0.20	Importance	-0.002	0.740	0.74 [0.31, 1.12]	$< 10^{-4}$
0.20	Random	0.143	0.423	0.28^{ns}	0.96
0.25	Importance	0.009	1.992	1.99 [0.93, 2.82]	$< 10^{-4}$
0.25	Random	0.619	1.603	0.98^{ns}	0.83
0.30	Importance	0.024	4.682	4.68 [2.12, 10.56]	$< 10^{-4}$
0.30	Random	1.837	4.683	2.85^{ns}	0.54

Table 14: Per-phase interaction index on GSM8K: Qwen2.5-7B-Instruct (n=50). Outputs truncated at `max_new_tokens`=2047; absolute values should be interpreted with this limitation. ‡Excluded: extreme II values from truncation artifacts render this condition invalid.

s	Strategy	$\widetilde{\mathrm{II}}_{high}$	$\widetilde{\mathrm{II}}_{low}$	Δ [95% CI]	p_{Bonf}
0.15	Importance	0.000	0.029	0.03^{ns}	0.39
0.15	Random	0.089	0.033	-1.58^{ns}	0.69
0.20	Importance	0.004	0.180	0.09^{ns}	0.62
0.20	Random	0.201	0.356	4.88^{ns}	0.57
0.25	Importance	0.009	1.064	1.14 [0.62, 1.66]	$< 10^{-4}$
0.25	Random	0.802	1.213	11.1^{ns}	0.39
0.30	Importance	0.023	2.140	4.62 [2.88, 6.92]	$< 10^{-4}$
0.30	Random‡	*excluded (data quality)*			

rising truncation (16% $\rightarrow$ 76%); non-truncated analysis (§O) reveals a dual-regime structure.

Table 15: MATH-500 accuracy under compound sparsity (importance-based eviction, `max_new_tokens`=8192). δ: accuracy change vs. dense. Trunc: truncation rate under sparse condition. DS: n=50 (dense truncation 16%); QW: n=100 (truncation $\leq$ 4% all conditions).

Model	s	Dense	Sparse	δ	Trunc
DS	0.15	74%	48%	-26 pp	32%
DS	0.20	74%	30%	-44 pp	56%
DS	0.25	74%	8%	-66 pp	76%
DS	0.30	74%	4%	-70 pp	76%
QW	0.15	46%	42%	-4 pp	$\leq$4%
QW	0.20	46%	36%	-10 pp	$\leq$4%
QW	0.25	46%	33%	-13 pp	$\leq$4%
QW	0.30	46%	19%	-27 pp	$\leq$4%

Failure mode divergence. DeepSeek and Qwen exhibit qualitatively distinct failure modes under compound sparsity. DeepSeek failures are *truncation-dominated*: compound sparsity is associated with verbose, repetitive chain-of-thought that exceeds the token limit (truncation rises from 16% at dense to 32% at s=0.15 and 76% at s=0.30, even with `max_new_tokens`=8192). Qwen failures are *wrong-answer-dominated*: outputs complete within the token limit (truncation $\leq$ 4%) but produce incorrect answers, indicating that compound sparsity disrupts reasoning quality directly. This divergence likely reflects training-paradigm differences: DeepSeek's distillation from DeepSeek-R1 produces extended CoT traces susceptible to truncation under degraded conditions.

Language switching under high sparsity. DeepSeek problems exhibit language switching to Chinese mid-generation at the following observed rates: 0% at dense, 4% at s=0.15, and 40% at s=0.30 (intermediate sparsity levels not yet measured).[4] This qualitative pathology—absent in Qwen—is a *symptom* of compound-sparsity-induced degradation rather than a cause of accuracy loss: at s=0.30, Chinese-language and English-language outputs achieve comparable accuracy

[4]Language switching rates were detected from `gen_snippet` fields in 2048-token evaluations, not from the independent 8192-token generations used in Table 15. Rates under the 8192-token budget were not re-measured.

(∼10%), indicating that the accuracy decline reflects pruning-induced reasoning corruption regardless of output language. The switching pattern is consistent with the model's Chinese-language pretraining corpus surfacing under degraded attention in distillation-trained models.

O Non-Truncated Sensitivity Analysis

With `max_new_tokens`=8192, Qwen truncation remains $\leq 4\%$ across all sparsity levels, so the full-sample accuracy decline ($\delta = -4$ to -27 pp; Table 15) reflects pure reasoning quality degradation without truncation confound.

DeepSeek exhibits substantial truncation (16% at dense to 76% at s=0.25–0.30). Restricting to non-truncated outputs reveals a *dual-regime* structure. At $s \leq 0.20$, non-truncated accuracy remains 68–71% (vs. 88% dense; n_{NT}=34 at s=0.15, n_{NT}=22 at s=0.20), while full-sample accuracy drops to 30–48%—indicating that degradation at low sparsity is primarily truncation-driven rather than reflecting reasoning quality loss. At $s \geq 0.25$, non-truncated accuracy itself collapses: 33% at s=0.25 and 17% at s=0.30 (n_{NT}=12 for both; reported as reference values given small samples), showing that high-dose compound sparsity degrades both generation length *and* reasoning quality simultaneously. The NT accuracy cliff from 68% (s=0.20) to 33% (s=0.25) localizes the transition between truncation-driven and dual-collapse regimes.

This dual-regime structure complements the importance discrimination collapse analysis (§6): at low sparsity, compound interaction primarily manifests as increased verbosity (degraded attention produces repetitive CoT); at high sparsity, the interaction additionally corrupts reasoning-critical computations.

P Benjamini-Hochberg FDR Correction

Table 16 compares Bonferroni (k=4, per model×strategy family) and Benjamini-Hochberg FDR correction across all 44 statistical tests. BH-FDR recovers 5 additional significant results beyond Bonferroni, all in low-effect-size conditions (random eviction or cross-training-paradigm random comparisons). All 32 Bonferroni-significant results remain significant under BH-FDR.

The 5 BH-only results are: DeepSeek random s=0.25 and s=0.30 per-condition Wilcoxon (p_{Bonf}=0.064/0.065, BH q=0.002), DeepSeek random s=0.25 → 0.30 adjacent-sparsity step (p_{Bonf}=0.078, BH q=0.002), and cross-training-paradigm random s=0.25/s=0.30 Mann-Whitney (p_{Bonf}=0.66/0.17, BH q=0.018/0.005). All involve random eviction—the weakest effects—indicating that Bonferroni is conservative but does not miss any importance-based findings.

Table 16: Bonferroni vs. BH-FDR correction summary (α=0.05, 44 tests). "BH only" denotes tests significant under BH-FDR but not Bonferroni.

Category	Bonf. sig.	BH-only sig.
Per-condition Wilcoxon (k=16)	10	2
Imp vs. rand (k=8)	8	0
Adjacent sparsity (k=12)	10	1
Cross-paradigm Mann-Whitney (k=8)	4	2
Total (k=44)	32	5

Q Minimum Phase Length Sensitivity

Table 17 reports the paired median difference Δ under three `min_phase_length` (mpl) settings across all 16 conditions (2 models × 2 strategies × 4 sparsity levels). Conclusions are robust: all importance-based conditions remain significant at $p_{\text{Bonf}} < 10^{-10}$ across mpl $\in \{5, 10, 20\}$; random conditions maintain the same significance pattern (non-significant at $s \leq 0.20$, significant at $s \geq 0.25$). Median Δ values vary by $< 5\%$ between mpl settings for DeepSeek importance, supporting that the severity gradient is not an artifact of the merging threshold.

The main effect of increasing mpl is reduced n_{eff}: at mpl=20, DeepSeek retains 433/500 problems and Qwen retains 289/500 (mid-null rate rises to 71.2% and 84.6%, respectively). Despite this sample reduction, all significance conclusions hold.

R Segmentation Robustness

Table 18 reports the paired median difference Δ under tercile, quartile, and quintile segmentation for both models under importance-based eviction. Phase gradient direction and statistical significance are preserved under quartile and quintile segmentation (n=50 spot check), indicating that the tercile-based findings are not an artifact of the segmentation choice.

The quartile and quintile Δ values are comparable to or slightly larger than tercile values because finer-grained segmentation isolates more extreme

Table 17: Sensitivity of median Δ to `min_phase_length` (mpl). n_{eff}: effective sample size after exclusions. All importance-based $p_{\text{Bonf}} < 10^{-10}$; "ns" denotes $p_{\text{Bonf}} > 0.05$.

Model	Strategy	s	mpl=5	mpl=10	mpl=20	$n_{\text{eff}}(10)$	$n_{\text{eff}}(20)$
DS	Imp	0.15	0.391	0.391	0.393	496	433
DS	Imp	0.20	1.215	1.215	1.235	496	433
DS	Imp	0.25	3.986	3.986	4.017	496	433
DS	Imp	0.30	8.481	8.481	8.420	496	433
DS	Rand	0.15	0.014	0.014^{ns}	0.012^{ns}	496	433
DS	Rand	0.20	0.031^{ns}	0.031^{ns}	0.023^{ns}	496	433
DS	Rand	0.25	0.257	0.257	0.190	496	433
DS	Rand	0.30	0.510	0.510	0.410	496	433
QW	Imp	0.15	0.066	0.066	0.066	475	289
QW	Imp	0.20	0.307	0.307	0.325	475	289
QW	Imp	0.25	1.074	1.074	1.106	475	289
QW	Imp	0.30	3.155	3.155	3.581	475	289
QW	Rand	0.15	-0.011^{ns}	-0.011^{ns}	-0.011^{ns}	475	289
QW	Rand	0.20	0.015^{ns}	0.015^{ns}	0.015^{ns}	475	289
QW	Rand	0.25	0.393	0.393	0.266	475	289
QW	Rand	0.30	0.913	0.913	0.788	475	289

Table 18: Segmentation robustness: median Δ under tercile (n=496/475), quartile (n=50 spot check), and quintile (n=50 spot check) segmentation. Importance-based eviction; all $p_{\text{Bonf}} < 10^{-7}$.

Model	s	Tercile	Quartile [95% CI]	Quintile [95% CI]
DeepSeek	0.15	0.391	0.491 [0.44, 0.70]	0.505 [0.33, 0.71]
	0.20	1.215	1.388 [1.05, 1.79]	1.197 [1.02, 1.76]
	0.25	3.986	4.322 [3.62, 8.06]	4.960 [3.80, 9.02]
	0.30	8.481	12.930 [9.83, 19.10]	11.548 [10.06, 18.51]
Qwen	0.15	0.066	0.226 [0.14, 0.27]	0.203 [0.13, 0.26]
	0.20	0.307	0.561 [0.35, 0.77]	0.517 [0.29, 0.75]
	0.25	1.074	1.870 [1.05, 2.44]	1.801 [1.13, 2.38]
	0.30	3.155	4.097 [2.91, 4.94]	4.009 [3.28, 5.34]

phases (the "lowest" quintile captures a more concentrated low-sparsity segment than the lowest tercile). The monotonic dose-response (Δ increasing with s) is preserved across all three segmentation schemes for both models, supporting that the severity gradient is not an artifact of the tercile segmentation choice.

S Robustness to Asymmetric Sparsity Allocation

Our main experiments use symmetric sparsity ($s_{\text{head}} = s_{\text{KV}} = s$). To verify that the phase-gated severity gradient does not depend on this symmetric assumption, we evaluate two asymmetric configurations at a fixed total sparsity budget equivalent to s=0.25, using importance-based eviction on DeepSeek-R1-Distill-Qwen-7B.

Both asymmetric configurations produce large positive Δ (+19.13 and +18.25), supporting that the phase-gated severity gradient is robust to the head/KV allocation ratio. The KV-heavy configuration (A) exhibits higher mid-phase II (6.05 vs. 3.97), suggesting that the KV eviction axis has a stronger impact on intermediate reasoning phases. Both asymmetric Δ values exceed the symmetric baseline (3.99), consistent with one axis operating at s=0.30—above the symmetric maximum tested—which amplifies overall severity. A systematic investigation of the full asymmetric interaction surface is left for future work.

Table 19: Asymmetric sparsity robustness: per-phase II under two head/KV allocations (DeepSeek, importance-based eviction, n=50 spot check).

Config	s_{head}	s_{KV}	$\widetilde{\text{II}}_{low}$	$\widetilde{\text{II}}_{mid}$	$\widetilde{\text{II}}_{high}$	Δ
A (KV-heavy)	0.20	0.30	19.51	6.05	0.38	+19.13
B (head-heavy)	0.30	0.20	18.77	3.97	0.52	+18.25
Symmetric	0.25	0.25	3.97	—	0.02	+3.99

T Outlier Robustness Analysis

The extreme mean/median ratio (up to 97,640:1 in individual conditions) reflects heavy-tailed II distributions driven by near-zero denominators in specific problems. We verify robustness via symmetric winsorization at 5% and 10% levels. All median-based statistics—paired median difference Δ, Wilcoxon signed-rank p-values, and amplification ratio G—are strictly invariant under winsorization, as expected for rank-based estimators. The five core claims—compound exceeds individual, importance exceeds random, monotonic dose-response, $G \gg 1$ for importance (auxiliary illustration of phase concentration), $G \approx 1$ for random—are fully robust. The extreme outlier severity further validates our choice of median-based inference over mean-based alternatives. G values in this robustness analysis follow the ratio-of-medians convention (Table 6); the primary G definition (Eq. 2) is similarly invariant as a median of per-problem ratios.

U <think> Token Phase Validation

DeepSeek-R1-Distill-Qwen-7B generates explicit <think>...</think> delimiters that demarcate chain-of-thought reasoning from final answers, providing a model-generated ground truth for reasoning phase boundaries. We evaluate whether our attention-sparsity-based phase segmentation (§3) recovers these boundaries on GSM8K (n=48 problems with valid phase assignments), using adaptive per-sequence quantile segmentation rather than a fixed global threshold.

Naming convention. A note on terminology: DeepSeek's <think> phase (active reasoning) exhibits *low* attention sparsity—attention is broadly distributed across many positions during exploratory computation. Conversely, the answer phase exhibits *high* attention sparsity—attention concentrates on a few positions during structured output. Thus PACS "low-sparsity phase" corresponds to the model's thinking phase, and "high-sparsity phase" to the answer phase. The physical phenomenon is consistent: broadly distributed attention characterizes reasoning in both frameworks.

Token-level agreement. We binarize each token as "thinking" (inside <think>) or "answer" (outside) and compare against the sparsity-based phase assignment. The adaptive segmentation achieves substantial agreement: κ=0.88 (per-problem mean 0.85, median 0.88), with token-level accuracy of 94.2%. Per-problem AUC-ROC is 0.98 (median 0.99), indicating near-perfect discriminability between thinking and answer phases based on attention sparsity alone. Of problems, 75% achieve $\kappa \geq 0.80$ and 100% achieve $\kappa \geq 0.60$. The <think> phase exhibits significantly lower attention sparsity than the answer phase (0.656 vs. 0.763; Cohen's $d = -0.80$), indicating that attention distribution breadth reliably distinguishes reasoning from structured computation.

Boundary proximity. The median distance between the sparsity-based phase boundary and the nearest <think>/</think> delimiter is 1 token, with 81.2% of boundaries falling within 10 tokens. This tight alignment indicates that the attention sparsity transition closely tracks the model's self-identified shift from exploratory reasoning to structured answer generation.

Phase composition and II correlation. We examine whether the fraction of thinking tokens within each phase correlates with compound severity (II). At the mid-phase level, the thinking-token fraction correlates with II (Spearman ρ=0.49, $p < 0.001$), consistent with mid-phase segments that contain more reasoning content being more vulnerable to compound sparsity. Low- and high-phase correlations are non-significant, as expected: these phases are compositionally dominated by thinking or answer tokens, respectively, leaving insufficient variance for correlation.

MATH-500 indirect validation. Direct κ validation requires explicit <think> delimiters, available only on GSM8K. On MATH-500 (n=500), we instead assess *temporal ordering consistency*: the fraction of problems where the low-sparsity phase precedes the high-sparsity phase, consistent with a thinking-then-answering generation structure. DeepSeek achieves a temporal ordering score of 0.576 with 100% of problems exhibiting the expected direction (low-sparsity before high-sparsity). Qwen, which lacks <think> delimiters, shows weaker but directionally consistent temporal ordering (0.479). This indirect evidence supports the validity of sparsity-based phase segmentation on MATH-500 but does not constitute direct κ validation.

Scope. The direct κ validation (κ=0.88, n=48) is limited to DeepSeek-R1-Distill-Qwen-7B on GSM8K; MATH-500 has only indirect (temporal

ordering) validation. Qwen2.5-7B-Instruct does not produce explicit reasoning delimiters, precluding direct κ validation on either benchmark. Phase segmentation is computed on the dense model's attention patterns and is independent of sparsity level, so the agreement statistics are reported once rather than per-s.

U.1 Non-Think Token Quality Under Compound Sparsity

Tokens generated outside `<think>...</think>` delimiters—i.e., the final answer tokens—fall predominantly in the high-sparsity attention phase, where compound interaction is minimal (§5.11: 93.2% of `\boxed{}` markers reside in the high-sparsity phase). We illustrate the differential impact of compound sparsity on non-think (answer) versus think (reasoning) tokens with two representative examples from the GSM8K validation set (DeepSeek-R1-Distill-Qwen-7B, s=0.25, importance-based eviction).

Example 1: Preserved answer, degraded reasoning. Under compound sparsity, the non-think answer tokens (`The answer is 42`) are generated with near-baseline perplexity (PPL ratio 1.2×), while the preceding think-phase tokens exhibit 4.8× PPL inflation. The final answer remains correct despite substantial reasoning-trace degradation—a "silent reasoning error" scenario where intermediate logic is corrupted but the output format masks the damage.

Example 2: Cascading failure at high sparsity. At s=0.30, reasoning-phase PPL inflation (11.3×) propagates into the answer phase: the model produces a syntactically valid but numerically incorrect `\boxed{}` output (PPL ratio 3.1× on answer tokens vs. 1.1× at s=0.15). This illustrates the sparsity threshold beyond which compound severity breaches the phase boundary, consistent with the accuracy cliff observed for Qwen at s=0.30 ($\delta = -27$ pp; Table 15).

Implication. These examples illustrate that pruning-induced compound severity is phase-selective: non-think tokens are relatively protected at moderate sparsity ($s \leq 0.25$) because they coincide with the high-sparsity attention phase where importance scores retain discriminative power. At higher sparsity (s=0.30), discrimination collapse (§6.1) erodes this protection, and compound severity spills into the answer phase.

V Amplification Ratio (G) Details

The per-problem amplification ratio $G^{(i)} = \text{II}^{(i)}_{low}/\text{II}^{(i)}_{high}$ (Eq. 2) provides an auxiliary illustrative perspective on the primary metric Δ. Table 20 reports G with bootstrap 95% CIs for all conditions. G serves as a visualization aid; primary statistical evidence for the phase-modulated severity gradient is provided by Δ and the associated Wilcoxon tests (§5.2).

The G values illustrate that phase-concentrated severity is associated with importance-based eviction (G=7.5–79.8× for DeepSeek, 2.6–36.9× for Qwen), while random eviction suggests weak modulation at $s \geq 0.25$ ($G \approx 1.1$–1.3; CIs exclude 1) and near-uniform severity at $s \leq 0.20$ (CIs span or touch 1). The monotonic increase in G with sparsity is consistent with the Δ dose-response in Table 2, but G should be interpreted as a descriptive ratio rather than an independent statistical test—the formal evidence rests on the paired Δ and associated Wilcoxon tests.

Table 20: Per-problem amplification ratio G (median of $G^{(i)}$) with bootstrap 95% CIs (B=10,000). G serves as an auxiliary visualization of phase concentration; primary statistical evidence for the severity gradient is provided by the Interaction Index Δ (§5.2). $G \gg 1$ suggests phase-concentrated severity; $G \approx 1$ suggests uniform severity across phases.

s	Strategy	G [95% CI] (DeepSeek)	G [95% CI] (Qwen)
0.15	Importance	**7.5** [4.1, 11.4]	n.d.†
0.15	Random	0.5 [0.4, 0.6]	0.7 [0.4, 0.8]
0.20	Importance	**17.9** [12.7, 26.3]	**2.6** [0.5, 5.1]
0.20	Random	0.8 [0.6, 0.9]	0.9 [0.7, 1.0]
0.25	Importance	**49.1** [40.1, 57.8]	**9.5** [5.2, 14.8]
0.25	Random	1.1 [1.0, 1.2]	1.3 [1.2, 1.4]
0.30	Importance	**79.8** [63.3, 96.5]	**36.9** [25.3, 49.1]
0.30	Random	1.1 [1.0, 1.3]	1.3 [1.2, 1.4]

† $\widetilde{\mathrm{II}}_{high} \approx 0$; ratio undefined ($G$ CI: $[-0.2, 0.9]$). Δ=0.07, p=8.7×10^{-15} supports $\mathrm{II}_{low} > \mathrm{II}_{high}$.